



Review article

A systematic review on the analysis of trace materials via Raman spectroscopy: Advancements and forensic implications

Samiksha Chauhan, Lovlish Gupta, Sweetey Sharma^{*,1}

LNJN NICFS, School of Forensic Sciences, National Forensic Science University, An Institute of National Importance, Ministry of Home Affairs, Govt. of India, Delhi Campus, Delhi 110085, India

ARTICLE INFO

Keywords:

Raman Spectroscopy
Raman micro-spectroscopy
Surface Enhanced Raman Spectroscopy
Forensic traces
Ink analysis
Paint analysis
Gunshot residue and Soil analysis

ABSTRACT

Raman spectroscopy, complemented by advanced techniques such as surface-enhanced Raman spectroscopy (SERS) and micro-Raman spectroscopy, is increasingly expanding its forensic applications in analysing trace materials. Forensic practitioners are adopting it more widely due to its simplicity, speed, non-destructive nature, and minimal or no sample preparation required. Despite certain limitations, such as relatively low sensitivity and significant fluorescence interference, Raman spectroscopy has been greatly enhanced by advanced techniques like SERS and Resonance Raman Spectroscopy, which effectively mitigate these issues and help unlock the full potential of Raman analysis. This review compiles extensive data from three major databases—Web of Science, Scopus, and Google Scholar—to demonstrate the utility of Raman spectroscopic techniques in the forensic analysis of trace samples, particularly in the examination of inks, dyes, gunshot residues, paints, and related substances. The study shows that Raman spectroscopy has significantly simplified the analysis of forensic traces, offering a promising approach for the on-site examination of a wide range of traces with improved efficiency and accuracy.

1. Introduction

The world has recently witnessed a COVID-19 pandemic that has led to tremendous changes in the quality of life for many individuals. While it was widely anticipated that crime rates would decrease due to widespread restrictions and reduced public activity, the crime rate remained steady between 2020 and 2021, largely as a result of stay-at-home mandates. The pandemic also brought notable shifts in criminal behaviour—altering the types of crime committed, methodologies used, and crime patterns [1].

In response to these evolving challenges, the criminal justice system continues to entail robust strategies to augment its reform efforts. Forensic science has always been one of the strongest pillars supporting the criminal justice system [2]. *It is a case-based, research-oriented, science-based endeavour to study traces—remnants of past activities, such as an individual's actions and presence—through their detection, recognition, recovery, examination and interpretation in order to understand anomalous events of public interest [3].*

To achieve this, forensic science integrates a range of analytical

techniques from multiple disciplines to perform quantitative and qualitative examinations of various forms of forensic traces. The entire process is maintained by proper chain of custody, from the initial entry to the crime scene to the final court proceedings. The chain of custody includes the proper collection, preservation, packaging, and secure transfer of traces found at the crime scene. However, maintaining the chain of custody from the crime scene to the transportation of different exhibits presents various challenges [4]. Furthermore, the quantity and quality of the traces recovered from the scene play a crucial role in the outcome and reliability of forensic science.

Sometimes, the recovery of trace material poses challenges for forensic examination [5]. Over the decades, various atomic and molecular spectroscopic techniques have been applied in forensic science. These non-destructive analytical techniques serve multiple purposes [6]. For example, ultraviolet spectroscopy is commonly used for forensic examination of ink [7]. Similarly, vibrational spectroscopy has been widely employed to analyse trace items, including fibre, inks, etc [8]. Researchers have also demonstrated the effective use of Raman spectroscopy to determine the composition of both burned and unburned

* Corresponding author.

E-mail addresses: samikshachauhan305@gmail.com (S. Chauhan), lovlishgupta95@gmail.com (L. Gupta), sharmasweetey511@gmail.com, sweetey.sharma_dc@nfsu.ac.in (S. Sharma).

¹ ORCID: 0000-0002-7886-4672

<https://doi.org/10.1016/j.forensiint.2025.112609>

Received 2 December 2024; Received in revised form 11 August 2025; Accepted 13 August 2025

Available online 15 August 2025

0379-0738/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

particles, such as gunshot residue, in their studies[9,10]. This technique provides rapid and efficient results with minimal or no sample preparation, offering advantages over other similar techniques [11].

The technique primarily relies on a phenomenon known as scattering of light, in which visible or monochromatic light is used to irradiate a sample. Upon interaction, the sample emits scattered photons. These photons transfer energy that leads to the vibration of the molecule to a virtual state, leading to the generation of Stokes and anti-Stokes lines in the Raman spectrum. Interpreting these spectra is one of the crucial step in spectroscopic analysis. However, spectral interferences- such as fluorescence production, matrix effects, and sample degradation due to analytical conditions- pose significant challenges, often requiring the use of supplementary analytical tools [12–14].

The fundamental principle of Raman spectroscopy is based on the “scattering phenomenon”, whereby electromagnetic radiation passing through a transparent medium interacts with all types of particles that constitute the material [15,16]. A key factor influencing this interaction is particle size: if the particles are smaller than the wavelength of the incident light, they undergo scattering [17]. However, most of this scattered radiation retains the same wavelength as the incident light, such radiations are called Rayleigh scattering, which occurs prominently due to elastic collision. On the other hand, a little fraction of the scattered radiation has a different wavelength from the incident radiation. This This inelastic scattering is referred to as Raman scattering and forms the basis of the Raman effect [18].

Raman Lines, which represent Raman effect, are characterized by the Raman shift, denoted by the Greek symbol delta nu $\Delta\nu$, and is denoted by the given formula.

$$\Delta\nu = (\tilde{\nu}_s - \tilde{\nu}_o) \text{ cm}^{-1}$$

Where:

$\tilde{\nu}_s$ = wavenumber of the incident radiation, and

$\tilde{\nu}_o$ = wavenumber of the observed scattered radiation/ lines.

The principle of Raman spectroscopy is governed by two major theories: Quantum (particle) theory and Classical (wave) theory” [19].

According to quantum theory, the interaction of radiation with matter is described in terms of collisions. These collisions can be elastic collisions, which involve no exchange of energy and inelastic collisions, in which energy is exchange, leading in transitions between quantized energy levels [20].

During inelastic collision, molecules gain energy (excitation) and undergo relaxation in their vibrational levels. These energy changes give rise to Raman scattering: **Stokes lines** are produced when a photon interacts with a molecule in its ground state, the molecule absorbs some energy and is excited to a higher vibrational level. Anti-Stokes lines are produced when a molecule is already in an excited vibrational state, it can release this energy by emitting a photon with a higher energy (shorter wavelength) than the incident photon [21]. A graphical representation of the Stokes, anti-Stokes line, and Rayleigh peak is represented in the spectrum given below in Fig. 1.

The Stokes and anti-Stokes lines in the spectrogram above are consistently observed across all the spectral ranges of Raman spectroscopy, appearing equidistant from the Rayleigh line and from each other. The Stokes line shows more intense peak compared to the anti-Stokes line, while the Rayleigh peak is the most intense of the three. According to Classical Raman theory, the interaction between the positive and negatively charged particles within a molecule and an external static electric field leads to charge separation, induction of a dipole moment, and polarization of the molecule [22].

A typical Raman spectroscopy instrument comprises five basic components; an intense laser source, a sample handling unit, a monochromator, a detector, and a signal processing unit or output device. Argon or krypton ion lasers are among the most commonly used laser sources in Raman spectroscopy. Unlike IR spectroscopy, Raman sample handling systems offer the advantage that glass containers can be used to

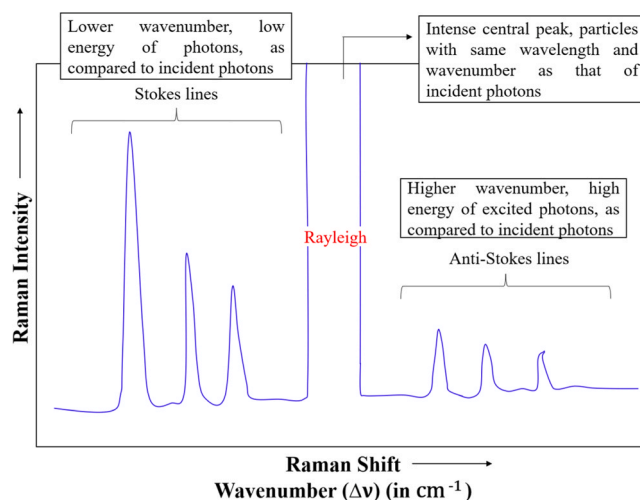


Fig. 1. Raman spectrum reflecting stokes, anti-stokes and Rayleigh peak.

hold aqueous solutions. One important feature of the setup is that the laser source is always positioned at a right angle to the detector. The most frequently used detector in Raman spectroscopy is the charge-coupled device (CCD) detector [23]. A basic block diagram for the Raman spectroscopy instrumentation is shown in Fig. 2.

The present review article explores the advancements in Raman spectroscopy for examining trace samples in forensic science from 2007–2024. The technique has been successfully applied in analysing various biological and chemical forms of material. It has demonstrated significant potential in both the preliminary and confirmatory examination of body fluids such as blood [24], semen [25] vaginal fluid [26] saliva [27], urine [28] and tears [29].

Raman spectroscopy is also effective in the detection and analysis of illicit drugs [30], their precursors, and new psychoactive substances [31]. The technique provides rapid and efficient discrimination of many cosmetic products, such as lipsticks [32,33], kajal [34] and nail paint [35]. In addition, several researchers have used Fourier Transform-Raman spectroscopy to differentiate various oils and edible fats without any sample preparation. It is also possible to determine the type and concentration of adulterants in various edible oils [36].

Raman Spectroscopy offers notable advantages in the examination of physical traces, such as paint in hit-and-run cases. Paint discrimination has proven useful in vehicle identification, especially in road accident investigations[37]. In ballistics, the technique is employed for the analysis of gunshot residue in cases involving firearms and related weapons. In the field of forensic physics, Raman spectroscopy has shown great potential in detecting and differentiating pigments and colorants, as well as in the analysis of ink and dyes for forensic purposes.

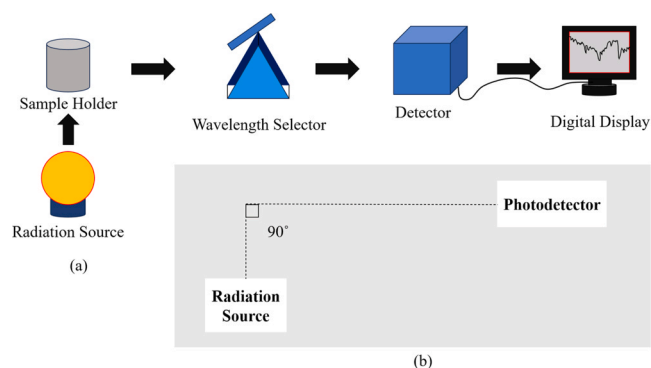


Fig. 2. (a) Block diagram for the Raman Instrumentation, (b) Arrange of Laser source and photodetector are shown perpendicular to each other.

Surface-enhanced Raman spectroscopy (SERS) and Spatially offset Raman spectroscopy (SORS) are two of the most widely used techniques in forensic trace analysis [38]. SERS involves the use of various substrates for the rapid detection and analysis of traces, particularly in the fields of agriculture and food forensic science [39]. The technique also incorporates the use of nanoparticles as substrates, for providing high efficiencies in analysing trace samples such as ink [40].

SERS offers several advantages, including enhanced sensitivity and selectivity over conventional Raman techniques, enabling the reliable detection of even minute traces recovered from crime scenes.

In the analysis of trace materials such as paint and fibre samples, SERS helps in identifying pigments, dyes and colorants. Fikiet et al., [41] demonstrated the application of SERS in detecting art forgeries, showing that the technique can identify extremely small amounts of information without compromising the integrity of the artwork. Further studies have confirmed the effectiveness of SERS in differentiating among various dye classes, including azo dyes, triarylmethane dyes, and lake dyes.

Moreover, SERS is valuable in identifying the dyes used to color fibres, which is crucial forensic trace material. Several methods have been developed to identify natural dyes in fibres using SERS without dye extraction. These approaches often utilise silver nanoparticles to enhance the SERS signal, allowing for dye identification directly on the fibre. This non-destructive approach is particularly important for preserving the integrity of the forensic sample.

Raman spectroscopy provides information about samples in the form of a spectrum, commonly referred to as a Raman spectrogram, which must be interpreted for further identification. Accurately interpreting Raman spectra to extract the most valuable information is a crucial step in forensic analysis [42].

Various statistical tools, such as chemometrics, play an imperative role in this analytical process by providing the necessary support for data interpretation. These tools include mathematical and statistical techniques to analyse the data obtained during the examination [43]. The combination of spectroscopy and chemometrics is widely regarded as ideal for chemical analysis in forensic science. With ongoing advancements in science and technology, this approach has become more significant and relevant than traditional methods of analysis. The chemometric process typically involves various steps, including data collection, pre-processing, multivariate analysis, calibration and data validation [44]. Pattern recognition methods used in chemometrics can be categorized as either supervised or unsupervised. Supervised methods include techniques such as unequal class model (UNEQ) and Discriminant analysis using Partial Least Squares Discriminant Analysis (PLS-DA), Support Vector Machine Discriminant Analysis (SVM-DA), Soft Independent Modelling of Class Analogy (SIMCA), Linear and Quadratic Discriminant Analysis (LDA and QDA). Unsupervised methods include Principal Component Analysis (PCA), K-means clustering, and Hierarchical Cluster Analysis (HCA) [8,43].

There is a significant need for holistic review and further research in the analysis of traces within forensic ballistics, forensic physics and questioned document examination (QDE). This review also includes a fundamental bibliometric analysis of research and review papers published in these domains concerning various types of forensic traces. The bibliometric analysis reveals that the majority of publications in the QDE domain focus on the analysis of different types of inks using Raman spectroscopy. However, only a limited number of studies have been published on the analysis of historical manuscripts within the same field.

To date, various forms and advancements of Raman Spectroscopy have been extensively applied across several domains and sub-domains of forensic science. Raman Spectroscopy has emerged as a versatile and indispensable tool, with wide-ranging applications. In forensic chemistry, it assists in the analysis of explosives, narcotics, and toxic substances [45]. In forensic biology, the technique supports the identification of biological fluids, tissues, and, cellular materials [46]. In forensic material science, Raman Spectroscopy is crucial for examining fibres, paints, and other physical traces [45].

Additionally, the technique has been employed in forensic toxicology for the detection of poisons and drugs, and in forensic document analysis for authenticating documents and detecting forgeries [47]. Advanced applications also extend to forensic anthropology, where Raman spectroscopy is used for the analysis of bones and teeth, and to forensic entomology, where it supports the study of insects to estimate the post mortem interval [48].

The widespread adoption of Raman Spectroscopy across these diverse forensic disciplines highlights its crucial role in enhancing the accuracy and efficiency of forensic investigations. Fig. 3 depicts the applications of Raman Spectroscopy in various domains and sub-disciplines of Forensic Sciences.

Numerous papers have also been published on the forensic analysis of paints and gunshot residues using Raman spectroscopy within the domains of forensic physics and ballistics. However, these publications are comparatively fewer than those focused on ink analysis, but more numerous than those related to the analysis of toners, dyes and other pigments. This bibliometric analysis specifically and explicitly focuses on publications related to the analysis of trace samples in forensic science using Raman spectroscopy, as illustrated in Fig. 4.

2. Types of Raman spectroscopy

Raman spectroscopy offers a wide range of capabilities for the detection, identification, and individualization of various Raman-active materials. Over time, many modifications and advancements in Raman spectroscopy have been reported in the literature. Fig. 5 represents the different Raman Spectroscopic techniques, while Table 1 summarizes these techniques along with their basic principles, advantages, and limitations.

3. Raman microscopy

Raman microscopy is one of the most impactful innovations in Raman spectroscopy and has been widely used as a non-destructive method for forensic analysis. Its primary application involves the *in-situ* detection of various traces [72]. Studies have reported that Raman micro-spectroscopy has also been applied to the analysis of textile fibres, owing to its high discrimination potential and identification power [72, 73]. In addition, the technique offers numerous benefits in the analysis of post-blast residues (PBR) and explosive materials. With its high



Fig. 3. Various Applications of Raman Spectroscopy in Forensic Sciences.

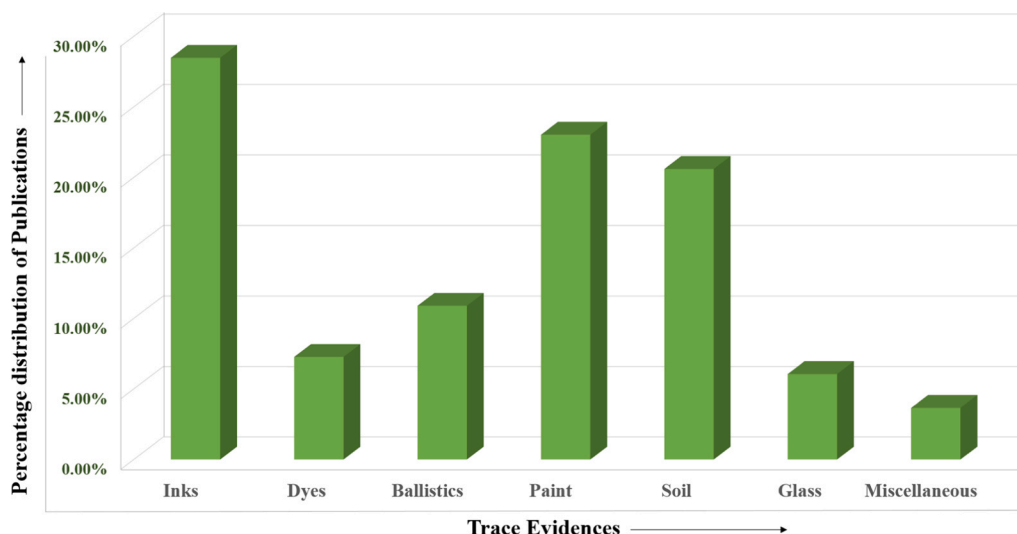


Fig. 4. Graph representing frequency of published articles on the analysis of different trace materials viz. Raman Spectroscopy.

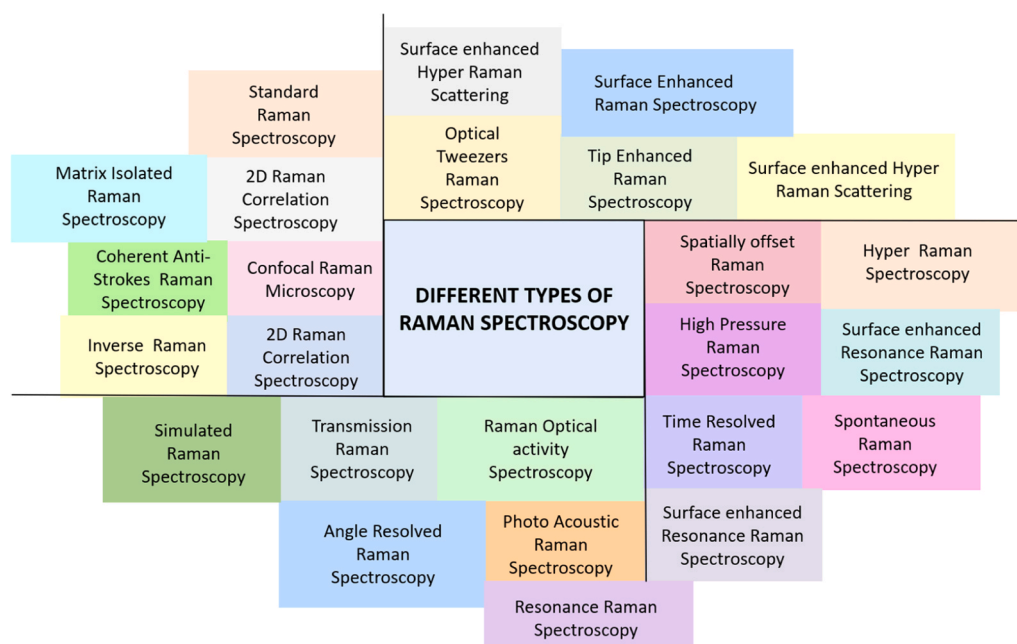


Fig. 5. Different types of Raman spectroscopy.

sensitivity and specificity, Raman microscopy enables the detection of microscopic particles from explosives such as Trinitrotoluene (TNT), Hexogen (RDX) and Pentaerythritol tetranitrate (PETN) as well as improvised explosive devices. Furthermore, Raman microscopy plays a vital role in analysing PBR, assisting investigators in reconstructing the composition of explosive devices [74,75]. Confocal Raman microscopy is a label-free, non-destructive imaging method used to examine various solid and liquid samples, including samples in variable environments. Raman microscopy is also closely related to fluorescence microscopy, making it a viable technique for comprehensive analysis of various forensic samples, including the imaging of cells and tissues [76]. The technique enables the analysis of the spatial distribution of the compounds, allowing for effective visualisation of samples. It has been reported that, compared to infrared microscopy, Raman microscopy is more suitable for analysing samples in aqueous environments [77]. Moreover, Raman microscopy has enabled Raman imaging by aligning imaging with the Raman spectra in a parallel configuration. Several

Raman tags have been reported in the literature, which have been significantly used to visualize smaller molecules within living cells. Therefore, Raman spectroscopy is considered one of the most prominent techniques for molecular imaging [78,79]

4. Analysis of inks, dyes, toners and pigments

Questioned document examination involves the analysis of handwriting, paper, and various types of inks in investigating fraud, forgery, and suspicious documents- such as suicide notes, counterfeit currency, and falsified educational or transportation documents. Handwriting analysis relies on the skills and experience of the document examiner. Currently, microscopic and spectroscopic techniques are widely used in document examination. Notably, Raman spectroscopy and its advancements- such as Surface-Enhanced Raman Spectroscopy (SERS) and micro-Raman spectroscopy (μ -Raman) have demonstrated significant promise in ink and toner analysis, offering high sensitivity, specificity,

Table 1

Different types of Raman Spectroscopic techniques, along with their basic principles, advantages, and limitations.

S. No.	Type of Raman Spectroscopy	Principle/ Theory	Advantage	Limitations	Ref.
1.	Standard Raman spectroscopy	The standard Raman spectroscopy relies upon the inelastic scattering of photons	–	Low cross-sectional sensitivity, interferences due to the fluorescence.	[18]
2.	SERS	The most sensitive Raman spectroscopy technique, enabling the structural detection of various analytes at lower concentrations. This greater sensitivity is achieved by amplification of electromagnetic fields, which occurs when the analyte of interest is in close proximity to nanostructured metal surfaces.	Significant signal enhancement. High intensity signal.	Needs tagging or immediate proximity to the molecule	[18, 49]
3.	Surface-enhanced hyper Raman Scattering	It relies upon the inelastic sum of frequency scattered from two subsequent photons. The enhancement results from molecular transition due to the total frequency of incident photons, corresponding to molecular electronic resonance.	Higher sensitivity than SERS. Ultrafast analysis of dynamic complex reactions	Requirement of complex complex-specific laser	[18, 50]
4.	Surface-enhanced resonance Raman spectroscopy (SERRS)	The technique involves the use of a chromophore present in the analyte with equal frequency to that of the excitation used to excite the plasmon.	Higher sensitivity and selectivity for the analysis of various analytes	Protein must be absorbed onto the SERRS active surface.	[51]
5.	Confocal Raman Microscopy	Works by focusing the laser light on the sample using the objective of the microscope.	Low fluorescence Detailed mapping. Non-invasive optical section of tissue, without dissection, or physical cut.	Requires careful hardware calibration and setup, which can be time-consuming	[52]
6.	Coherent Anti-strokes Raman Spectroscopy (CARS)	It is a non-linear technique, which enhances Raman signals, by using coherent laser beams to intensifies higher frequency than excitation frequency. The major principle relies upon multiple excitation laser sources, which include a coherent monochromatic field and a Stokes excitation field, overlapping spatially and temporally.	Low fluorescence	Needs specifically tuned excitation wavelength	[53]
7.	Photo Acoustic Raman Spectroscopy	The technique uses low power continuous wave laser excitation of the gaseous samples to be analysed, by the use of pulsed laser excitation.	Trace detection of gaseous mixture. Analyse pure rotational transition in gases.	Reduction of stray light is essential.	[18, 54]
8.	Angle Resolved Raman Spectroscopy	Technique involved a significant number of spectra with different polarization angles of both incident as well as scattered lights. The major principle involves separation of peaks by considerable symmetries of Raman tensor.	Provide information regarding the molecular alignment	Requirement of experimental set up	[55]
9.	Two-dimensional Raman correlation spectroscopy	It is a technique involving two independent variables i.e., wavelength and, frequency as a function to plot spectral intensities on two orthogonal axes.	Gives both synchronous and asynchronous spectra.	Temperature must be maintained to reduce temperature-induced variation of spectra.	[56]
10.	High pressure Raman spectroscopy	Confocal Raman imaging is the major principle behind working of High-Pressure Raman Spectroscopy.	Very high selectivity, Higher fingerprint analysis of materials.	Chemically reactive material analysis should be monitored.	[57, 58]
11.	Transmission Raman spectroscopy	Transmission Raman occurs when light passes/ scatters from turbid materials, the materials do not absorb or get blocked by the light.	Non- invasive and non-destructive analysis	Limited to transparent and semi-transparent material	[59]
12.	Hyper Raman spectroscopy	Modified Raman Spectroscopy includes three photons processing that analogue the Raman Scattering. Out of the three, two pump photons get converted into one photon of Raman scattered light and one phonon.	Can provide vibrational information on molecules.	Raman Scattering is suppressed because of non-Raman-active modes.	[60]
13.	Resonance Raman Spectroscopy	Spectroscopic instrument which tends to measure the shift in frequency of photon at the time when energy required for electronic transition equals energy of photons from incident light.	Significant signal enhancement	High fluorescence Needs specifically tuned excitation wavelength	[61]
14.	Spatially offset Raman spectroscopy	Technique works upon scattering or illuminating properties of certain objects, that they are neither completely transparent to light nor completely blocks it. This results into inelastic scattering due to Raman effect.	Highly accurate chemical analysis.	Total signal reduces with increase in offset signals	[62]
15.	Time resolved Raman spectroscopy	Techniques useful in monitoring the frequency of different vibrational modes, it monitors the changes occurring in the intensity of the peak, as per the time. Also helps in studying dynamics of vibrational modes.	Helps in extracting the data related to structural, kinetic and molecular interaction	Limited sensitivity	[63]
16.	Stimulated Raman Spectroscopy	The basic principle involves the incidence of two photons, which are the source of two laser beams which simultaneously coincides the sample.	Background imaging map of molecules is provided	Needs specifically tuned excitation wavelength	[18, 64]
17.	Matrix Isolated Raman Spectroscopy	Techniques useful in estimating vibrational modes of molecules. In this technique, host matrix such as a noble gas, Solid matrix, frozen solvent, etc. are used to trap individual molecules. The molecules are isolated within the matrix capturing it and can then be excited using suitable light source. And undergoes electronic and vibrational transitions after absorbing photons.	Analyse reactive species, Enhanced spectral resolution	Matrices interfere the analysis. Limited matrix choice is one limitation, Low sensitivity.	[65]

(continued on next page)

Table 1 (continued)

S. No.	Type of Raman Spectroscopy	Principle/ Theory	Advantage	Limitations	Ref.
18	Raman optical activity spectroscopy	The instrument principally measures smaller differences in the intensity of Raman scattering from chiral molecules specifically in circularly polarised light, in right and left direction.	Better analysis of chiral compounds. Unique identification of molecular structure and dynamics.	Not specifically identifies non-chiral compounds.	[66, 67]
19	Optical tweezers Raman spectroscopy	Technique involves precision of optical tweezers to trap and manipulate microscopic particles with the analytical power of Raman spectroscopy to probe their molecular composition.	Non-destructive, label free analysis of trapped particles.	Low sensitivity, Photodamage potential increases due to intense laser beams.	[68]
20	Inverse Raman spectroscopy	The technique utilizes two laser beams i.e., a fixed frequency pump beam and another is a low frequency Stokes beam. When these beams interact with a sample, energy is transferred from the pump to the Stokes beam through stimulated Raman scattering, amplifying the Raman signals.	Higher sensitivity, better characterisation of molecular vibrations.	Higher cost, complex instrumentation, Background noise due to amplification.	[69]
21	Spontaneous Raman spectroscopy	When a photon incidence and interacts crystal molecule, it scatters both elastically or inelastically. Inelastically, the molecule is left in modified vibrational state. And the transfer of energy creates a phonon in lattice	High sensitivity, non-invasive technique, and inhibits multiphoton effects.	Weak strength signal	[70]
22	Tip enhanced Raman spectroscopy	Technique combines the use of both scanning probe microscopy along with Raman Spectroscopy. Three types of illumination are necessary including, reflection, side and top illumination, which are based on surface plasmon resonance and electrostatic lighting rod effect.	Higher sensitivity Excellent resolution. Better qualitative analysis.	Need of sharply focused illuminating laser.	[18, 71]

and non-destructive testing capabilities [80,81].

In order to examine forged documents, counterfeit currencies, and differentiate between genuine and fake documents, ink analysis is a crucial factor in forensic document examination. The identification of the type of ink and its constituent components provides valuable information regarding the age and authenticity of the documents. Inks such as carbon ink, fountain pen ink, and ball point ink are commonly used for writing. These inks comprise synthetic dyes and other materials such as pigments, resin, and solvents [82]. Raman spectroscopy has been extensively used to analyse the synthetic dyes present in ink. A study on ink reported that SERS performed slightly better with 633 and 785 nm lasers in the analysis of ballpoint ink when compared to conventional Raman spectroscopy. Although the study could only determine a single pen ink, the scope for improvement with a wider analytic approach has been seen significantly in later years [83].

On the other hand, the analysis of toners has also been conducted alongside ink analysis in cases involving printed documents specifically by laser printers. Toners differ from inks in that they are used in laser printers, whereas inks are used in inkjet printers. In a research study, experts examined a total of 30 toner samples using Raman spectroscopy, and 90 % of toners were accurately identified with the help of spectral library. Although, the sample size was limited, the study demonstrated the potential of vibrational spectroscopy in examining color- photocopied documents.

Crimes involving counterfeit currency and bank notes are not new. As new counterfeiting methods have emerged, analytical techniques and detection methods have also advanced. Micro-Raman spectroscopy (μ -Raman) is a technique capable of identifying toners and their sources used in the printing counterfeit bank notes [81,84]. Additionally, the characterization of inks to discriminate between counterfeit and authentic banknotes using Raman Spectroscopy coupled with chemometric methods has shown significant potential [85].

Furthermore, studies have shown that Raman spectroscopy provides better results for analysing colour- printed document. Due to its non-destructive nature, it remains the first choice for analysing printer inks. However, a limitation of conventional Raman spectroscopy is its low signal strength caused by fluorescence. As a result, SERS has often been chosen for the detection and identification of ink in most cases. This technique quenches fluorescence, which typically causes

interference with the Raman signal [86]. SERS-based detection is advantageous due to its low cost, rapid analysis and high sensitivity for dye analysis [87].

Video spectral comparator (VSC) and thin-layer chromatography (TLC) are among the most popular techniques for document examination. VSC focuses on deciphering indentations, whereas TLC is primarily used for ink analysis. Scientists have used these techniques to examine liquid and gel inks by coupling them with Raman Spectroscopy and SERS. The VSC and Raman Spectroscopy turned out to be complementary techniques, with the latter exhibiting comparable discriminatory power for liquid inks to VSC [88].

Another important consideration is the paper material, which can interfere with ink analysis. Confocal Raman spectroscopy addresses this problem and offers high discriminatory power in characterizing blue ball point ink and seal inks on paper. Unlike FTIR, it is a much simpler and faster approach for identifying ink found in Chinese markets [89, 90]. Even conventional Raman spectroscopy has been used to identify and classify inks. In a study on ball point ink, almost 90 % of all the blue and black ballpoint ink samples were identified and differentiated based on Raman spectra alone [91]. Before 2016, major studies on Raman spectroscopy focused on black or blue ink in forensic document analysis. In 2016, red and green ballpoint ink were also examined using Raman spectroscopy and four other techniques. However, since, the discriminatory power of this technique was not adequate for the analysis, improvements were made in the following year [92].

Estimating the age of writing inks to determine the document age is another important aspect of questioned document examination and is commonly included in almost every ink analysis protocol. Traditionally, TLC and gas chromatography (GC) are popular techniques used for ink dating. Dye and pigments are two important elements of inks that need to be analysed to determine the age of the documents. Experts primarily focus on the degradation of dyes, pigments, and colorants over time to determine the ageing process. Researchers have found that Raman spectroscopy has the potential to estimate the age of papers for up to 15 years based on the decomposition of the dyes [93]. Further, scientific research has investigated the ageing process and heating time of the dye. The obtained Raman spectra were compared with standards, and the results supported previous studies. Although fluorescence remains a problem, the development of new methodologies may be possible based

on these observations.

Determining the crossing of ink lines is another important aspect of forensic document examination. Borba et al. [94] applied confocal Raman spectroscopy in four cases to determine crossed ink lines and obliterated lines. In conjunction with chemometrics, this technique identified the order of strokes and the nature of inks used in all the cases. Raman spectroscopy can analyse the intersecting strokes of inks and toners [95]. Another study examined 56 crossing lines drawn from a total of eight blue and black inks, all from different brands. The obtained spectra were analysed using chemometrics techniques, i.e., Raman hyperspectral imaging and Multivariate Curve Resolution-Alternating Least Squares (MCR-ALS), which accurately determined the chronological order of more than 70 % of crossings. This approach, in which different techniques were combined and applied together, produced better results in examination than microscopic analysis, according to further studies [96].

With time, the discriminatory power of Raman Spectroscopy and SERS has also increased in the analysis of pen inks. Different chemometric techniques have also been applied to enhance the performance of Raman spectroscopy. In a recent study [97], 140 samples of black gel pen ink were successfully examined by Raman Spectroscopy coupled with PLS-DA and the Uniform Manifold Approximation and Projection (UMAP) model. Currently, the techniques of Raman spectroscopy and SERS are improving at a faster pace. Hence, these techniques can be used to analyse blue and black ball and gel writing inks when applied with chemometrics approaches. The forensic analysis of ink or dye is multi-disciplinary and multi-dimensional in itself. Raman spectrometers have been widely used over the years for this purpose.

Raman spectroscopy, combined with chemometric techniques like PCA and PLS-DA, has proven effective in classifying inks on forged documents and analysing pigments in painting specimens. The technique has also been successfully used to identify and discriminate various organic pigments, their elemental patterns, and classes using Normal Raman Microscopy and SERS [97,98]. SERS has enabled the identification of over 30 different hair colorants with high accuracy, regardless of type, brand, or colorant. Electrochemical Surface-enhanced Raman Spectroscopy (EC-SERS) has enhanced the analysis of natural pigments present in lakes. At the same time, silver nanoparticles (AgNP) on cost-effective screen-printed electrodes (SPEs) have improved the SERS response of phenolic compounds [97,99,100]. Fig. 6 represents the distribution of research work published for the analysis of inks, dyes and pigments using Raman spectroscopy in various literature from 2007 - June 2024.

Raman Spectroscopy has also been applied to the non-invasive and non-destructive analysis of textile dyes in heritage textiles and the chemical analysis of powdered and azo dye molecules. Furthermore, integrating chromatography and Orthogonal Partial Least Square

Discriminant Analysis (OPLS-DA) with μ -Raman spectroscopy has enabled real-time detection and identification of stamp inks based on their unique molecular signatures [101]. The description of various analytical methods depicted in the literature for the analysis of inks, pigments, dyes and other traces from 2007 - June 2024 is highlighted in Table 2.

5. Analysis of gun-shot residue

Forensic ballistics includes the examination of firearms, ammunition, and injuries caused by such firearms. Firearms are used in cases of homicides, robberies, assaults, and other violent crimes or crimes where the intention of causing grievous harm is dominant. Firearm-related traces may be found at the crime scene and help in establishing links among the victim, suspect and crime scene. It follows the principle of comparison and the law of mutual exchange, where the exchange of gunshot residue (GSR) takes place from the culprit to the crime scene and the victim, their clothing or surroundings and later can be compared to link them to each other. A projectile can also be linked to its original firearm from which it has been fired [150].

As the name suggests, gunshot residues or firearm discharge, are composed of organic and inorganic components contained in the primer charge, propellants, and other ammunition parts. The organic part includes the analysis of gunshot residues, which yields valuable information related to the firearm, the manner of death, the range of firing, and injuries [151]. Traditionally, the analysis of GSR includes chemical tests and instrumental techniques for routine purposes. These methods detect and analyse the organic and inorganic constituents of GSR. The organic residues include compounds such as nitro glycerine, diphenylamine, etc. The basic metallic elements in the inorganic part of GSR involve barium, antimony, lead from barium sulfate, antimony nitrate, antimony sulfide. Inductively Coupled Plasma Mass Spectrometry (ICP-MS), Neutron Activation Analysis (NAA), Atomic Absorption Spectroscopy (AAS) and Scanning Electron Microscopy with Energy Dispersive X-ray Spectroscopy (SEM-EDX) are efficient in the characterization and elemental analysis of inorganic compounds [152]. As there is no standard protocol for the analysis of GSR, organic constituents can be examined using chromatographic techniques such as GC, TLC, and HPLC (High-performance chromatography) [153]. Vibrational spectroscopy also targets the organic compounds present in GSR, which can help determine the type of ammunition used. Raman spectroscopy easily and rapidly detects these components. Raman spectroscopy, however, is a screening tool for tracing GSR. Near Infra Raman spectroscopy and Confocal Raman micro-spectroscopy are suitable for the differentiation of GSR when applied with chemometrics methods as presented by various studies [154]. Raman spectroscopy can discriminate and characterise GSR in combination with techniques such as FTIR (Fourier transform infrared spectroscopy) and SEM/EDX [155]. Statistical methods can further analyse the obtained data. Raman micro-spectroscopy, accompanied by statistical analysis, identifies the inorganic and organic components of GSR collected by adhesive tapes. The detection limit of this method is low. It can estimate the GSR particles up to the size of 3.4 μm [9]. Developing lead-free ammunition has resolved the health and safety issues associated with lead. Consequently, the analysis of GSR was challenging because of the limited and difficult techniques available at that time. Lopez and his co-workers [156] studied the macroscopic particles of GSR and the effect of their interaction with blood in the cloth material using Raman imaging. They used different types of ammunition, the most significant of which was lead-free. The characterization and identification of GSR particles from non-toxic ammunition were achieved from the SEM-EDX stubs. Further, macroscopic smokeless powder was detected using SERS by the same research group. This time, the potential of SERS for detecting GSR particles was explored on cloth [157].

Many research papers have been published in recent years exploring the application of Raman micro-spectroscopy coupled with multivariate

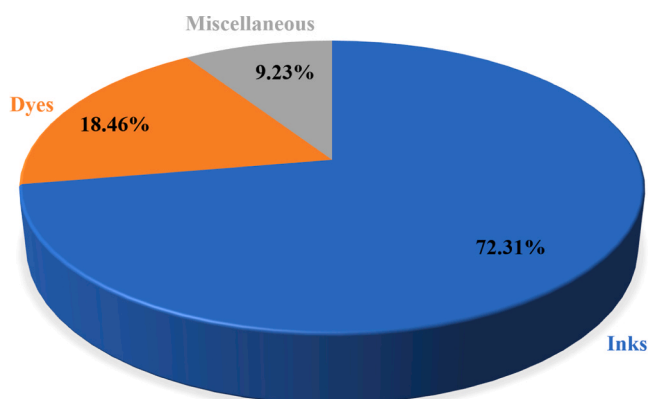


Fig. 6. Percentage distribution of Publications (2007-June 2024) specific to analysis of inks, dyes and pigments viz. Raman Spectroscopy.

Table 2

Available literature review on the forensic analysis of ink using Raman spectroscopy.

Category	Aim	Results	Year	Ref.
Inks	Examination of forensic ink samples	It is a non-destructive analytical method that enables the analysis of the minimal number of samples	2008	[102]
Pigments	Chemical analysis of modern organic tattooing inks and pigments	The isolation process has clearly distinguished dark and lighter-coloured pigments.	2008	[103]
Ink	Identification of iron-gall inks with near-infrared Raman micro spectroscopy	Characteristic Raman spectra for iron-gall inks were obtained successfully, and also NIR Raman has demonstrated paper documents	2008	[104]
Dyes	Analysis of synthetic dyes found in ballpoint pen inks	Highly qualitative dye spectra have been obtained from pure dye samples with NR spectroscopy; SERS has provided signal enhancement and fluorescent quenching.	2008	[105]
Dyes	Evaluation of trace materials and the Examination of Questioned Document	SERS is a highly valuable and precise technique for examining organic colorants of inks.	2008	[106]
Toners	Forensic analysis of colour toners by Raman spectroscopy	Raman is helpful in the detection of toner pigments, whereas FTIR was sensitive to the toner resin	2009	[84]
Dyes	Analysis of black dyes on textiles	The technique is non-destructive for identifying of dyes and their mixtures and provides rich spectroscopic signature.	2010	[107]
Pigments	Characterization of some orange and yellow organic and fluorescent pigments	Results suggested that different peaks were obtained for different pigments.	2010	[108]
Ink	Forensic analysis of black and color inkjet printer documents	The combination of 3 techniques (Raman spectroscopy, Laser Desorption Mass Spectrometry (LDMS) and Matrix Assisted Laser Desorption Ionization -Mass spectrometry (MALDI-MS)) have highly discriminated inkjet-printed documents.	2012	[86]
Ink	Analysis and classification of pigments and inks	Laser Induced Breakdown Spectroscopy (LIBS) and Raman spectroscopic techniques have shown advancement over chromatographic techniques due to no sample preparation and successfully	2012	[109]

Table 2 (continued)

Category	Aim	Results	Year	Ref.
Toners	Identifying a common origin of toner printed counterfeit banknotes	classified pigments and inks. Micro-Raman spectroscopy can be successfully applied to analyse of coloured toners used in printing counterfeit banknotes.	2012	[81]
Ink	Red ink entries of seals by Raman and Fourier transform infrared spectrometry	The technique has provided a non-destructive approach for the discrimination of ink pastes or oils of seals on paper.	2012	[110]
Ink	Analysis of stamped inks in questioned documents analysis	Complementarity of Raman and High-Performance Thin Layer Chromatography (HPTLC) has been used to classify samples into different groups based on ink compositions.	2013	[111]
Liquid and gel inks	Examination of liquid and gel inks	The technique is adequate for maintaining the integrity of ink traces to ensure chain of custody.	2013	[88]
Dyes	In situ analysis of ink dyes	The technique proves to be a more effective technique and can successfully be employed for in situ analysis of various ball-point pen inks	2013	[112]
Ink	Discrimination between authentic and counterfeit banknotes	Raman spectroscopy, PCA, and PLS-DA analysis have significantly allowed the determination of genuine and fake real notes with ease	2013	[85]
Ink	Analysis of Gel Pen Ink Writings on White Office Paper	Spectral analysis helped in discriminating between different classes of gel inks. Similar patterns were obtained for a range of inks.	2013	[113]
Ink	Forensic colour inkjet printer inks analysis	The technique allows the direct characterization of inkjet printer inks on paper. Raman feasible colorants were easily identifiable	2014	[114]
Ink	Variability in the Raman signature of writing pen inks	Discrimination of pens inks of different brands were performed	2014	[115]
Ink	Discrimination of blue ballpoint pen inks	Raman spectroscopy has significantly removed all the interferences that IR spectroscopy was not able to do and successfully discriminated selected samples.	2014	[90]

(continued on next page)

Table 2 (continued)

Category	Aim	Results	Year	Ref.
Ink	Discrimination of seal inks used for seals	Popular pigments such as Scarlet Powder, Pigment Yellow 55, Bronze red C and PbCrO ₃ (Lead II chromate) were successfully identified.	2014	[89]
Ink	Microanalysis of organic colorants	Results suggested that visual observations were more significant in most conditions, even by optical microscopy	2014	[116]
Ink	Forensic discrimination of inkjet-printed lines	This spectrometer could discriminate minute spectral differences between complex printing ink samples, even at low resolution.	2014	[117]
Ink	Analysis of blue ballpoint pen inks	The technique successfully differentiated different pen inks into complexes	2015	[118]
Ink	Determining the sequence of blue pen ink crossings	Combining Raman imaging with other analytical and chemometric techniques is an excellent futuristic tool for crossing over ink analysis	2015	[119]
Blue inks	Differentiation of blue writing inks	Raman spectroscopy and SERS have allowed the identification of inks.	2015	[120]
Dyes	Detection and identification of hair Dyes	SERS is a valuable technique in distinguishing various commercial brands of various hair dyes.	2015	[121]
Dye	Forensic Analysis of Colored Acrylic, Cotton, and Wool Textile Fibers	Successfully detected and analysed various acrylic colours.	2015	[122]
Ink	Identification purposes for ballpoint pen inks.	It was a non-destructive approach and has successfully discriminated ballpoint pen inks on paper.	2016	[91]
Historical manuscript	Analysis of a Visigothic historiated manuscript using RS	The entire analysis was <i>in situ</i> , where a microscope and Raman spectroscopy hyphenated with a CCD detector turn is more comprehensive for the analysis	2016	[123]
Ink	Forensic Analysis of Black, Blue, Red, and Green Ballpoint Pen Inks	Raman and FT-IR have become important tool for QD examination because of enhanced peaks in the analysis.	2016	[92]
Inks	Studying Raman spectra of ballpoint pen inks	Combining Raman spectra with chemometrics technique such as PCA is superior to the conventional direct manual examination of ball point pen inks.	2017	[124]

Table 2 (continued)

Category	Aim	Results	Year	Ref.
Ink	Analysis of the Aging Processes of Writing Ink using Raman Spectroscopy	Raman spectroscopy can either be used separately or coupled with GC, and is a prominent tool for the identification of the age of writing inks.	2017	[125]
Dyes	Document analysis using Raman spectroscopy and fluorescence analysis.	Raman spectroscopy is an effective tool for studying the ageing processes of the writing inks and the documents.	2017	[126]
Dyes	Tracking photo-degradation of triarylmethane dyes	SERS was reliable in the investigation of photodegradation of dyes, such as triarylmethane dyes: Basic fuchsin (the three homologues Rosaniline, Pararosaniline, and New Fuchsin), Methyl violet, and Crystal Violet.	2017	[127]
Ink	Differentiation of printing inks	Raman micro spectroscopy has the great potential to demonstrate the manufacturer of yellow toners and magenta inkjets, by distinguishable spectra.	2018	[128]
Inks	Source Determination of Red Gel Pen Inks	The technique possesses greater feasibility for subsequent analysis and determination of red gel pen inks.	2018	[129]
Ink	Discrimination of inkjet printer inks	Raman spectroscopy serves is a potent tool for differentiating of ink samples printed with inkjet printers.	2018	[130]
Ink	Investigation of dye content in commercial felt tip pens inks.	SERS coupled with UV/vis serves as sensitive and effective tool for the characterization of fugitive dyes.	2018	[131]
Ink	Characterization of Brazilian banknotes	Effective tool in characterising bands of phthalocyanine and diacrylide pigments from the inks. Genuine and counterfeit banknotes spectra showed similar bands, so, the multivariate analysis was further used and satisfactory results were obtained.	2019	[132]
Inks	Forensic Ink Analysis	Raman analysis and chemometric procedures have enabled correct classification between blue ink samples obtained from different countries.	2019	[133]
Ink	Identification of dye content, analysis of blue and black inks,	SERS were used as an effective tool for investigating the	2019	[134]

(continued on next page)

Table 2 (continued)

Category	Aim	Results	Year	Ref.
Ink	and degradation processes Analysis of sequence of intersecting strokes from laser toner and water-based ink analysis	degradation process of blue writing inks. Raman spectroscopy with a digital microscope has been proven to be a master technique in determining order of intersecting strokes.	2019	[95]
Ink	Determination of chronological order of interesting ink lines	Microscopic and Raman hyperspectral imaging proved significantly helpful for the detection of crossing strokes.	2019	[96]
Ink	Discrimination of blue ballpoint pens inks	Raman imaging spectroscopy showed great potential for the studying forensic samples, specifically document forensic examination.	2019	[135]
Ink	Raman spectra for dating documents and inks	Creating and using of a single digital database of spectral data of samples for comparative research further will allow objectifying the results of forensic examination of documents and inks.	2020	[136]
Inks	Discrimination of pen inks on document by SERS fabricated by magnetron sputtering	Reproducible thin film-based SERS substrates, have been prepared by a DC magnetron sputtering for the effective classifications of the ballpoint pen inks.	2020	[137]
Blue gel pen inks	Raman Spectroscopy in Forensic used for the Analysis of Blue Gel Inks on Paper	The combination of Raman Spectroscopy with chemometric analysis with the help of Pearson's correlation and PCA is proven to be a resourceful determination of the blue gel inks.	2020	[138]
Ink	Discrimination of inkjet, laser and photocopier printed documents	The proposed method is so relevant in actual forensic casework, and is evident in the determination of the source of unknown printed samples.	2021	[139]
Ink	Classification of Raman patterns of inkjet printer inks	When all three printed colour dot components were analysed, better performance of the combination of these techniques was most significant.	2021	[140]
Black gel inks	Discrimination and identification of source of inks	PCA and PLS-DA models were used along with Raman Spectrometry, proved to be a better tool for the classification of inks on forged documents.	2022	[97]
Red Gel ink	A study conducted using Self-organizing feature maps (SOFM)	SOFM and chemometric tools, including of PLS, have	2022	[141]

Table 2 (continued)

Category	Aim	Results	Year	Ref.
	for distinction and source identification of ink.	been suitable for depicting the discriminating analysis using non-destructive Raman. (100 % prediction).		
Synthetic organic pigments	Identification of different organic colorants by Raman microscopy, and SERS	A combination of Normal Raman Microscopy and SERS successfully attenuated to discriminate various organic pigments, their elemental pattern and classes.	2022	[142]
Pigments	Characterisation of pigments in ancient manuscripts and paintings	Raman Spectroscopy has been a fundamentally used to characterise of authenticity of pigments in painting specimens.	2022	[99]
Hair Dyes	Analysis of Hair Dyes using SERS	Hair dyes accompany different colorants. SERS can accurately identify more than 30 hair colorants, regardless of the type, brand and colorant used.	2022	[143]
Polyphenolic Dyes	Determination of polyphenolic components in natural lake pigments.	Electrochemical Surface-enhanced Raman Spectroscopy has been an insignificant tool in the analysis of natural pigments present in lakes. Silver nanoparticles (AgNP) on cost-effective screen-printed electrodes (SPEs) enhance the SERS response of phenolic compounds. This enables sensitive detection and identification of natural dyes.	2022	[144]
Historical synthetic dyes	Raman Spectroscopic approach for structural characterisation of ancient synthetic dyes.	Non-invasive and non-destructive analysis of several dyes, including textile dyes from heritage textiles, has been prominent. Raman Spectroscopy has also been indispensable in chemical analysis of both powdered and azo-dye molecules.	2023	[145]
Blue Inks	Analysis of ink on suspected documents	Raman Spectroscopy, associated with chemometrics, such as PCA, have been successful in characterisation of 45 different blue-ink pens.	2023	[98]
Synthetic dyes	Analysis of composition of synthetic dyes in ancient inks, their degradation patterns, to preserve cultural	Raman Spectroscopy and TLC-SERS have been analysed successfully to effectively demonstrate and discriminate the	2023	[100]

(continued on next page)

Table 2 (continued)

Category	Aim	Results	Year	Ref.
Blue Ballpoint pen ink	heritage (19th –20th century)	components in mixtures of organic pigments.	2023	[146]
	A case reported presented for analysis of related a blue ball pen ink and its related dating in a forensic questioned document.	Blue ballpoint pen inks often contain crystal violet, a triarylmethane dye, which produces a distinct Raman spectrum with intense peaks at 749 cm ⁻¹ and 1542 cm ⁻¹ , serving as a unique "molecular fingerprint". Therefore, Raman shifts can be utilized to estimate relative intensities of peak, for the dating of the relative documents.		
Stamp inks	Identification of different stamp inks using Raman and chromatographic analysis.	By integrating chromatography and (Orthogonal Partial Least Square Discriminant Analysis) OPLS-DA with micro-Raman spectroscopy, researchers can achieve real-time detection and identification of stamp inks based on their unique molecular signatures.	2023	[101]
Iron Gall inks and Logwood inks	Analysis and discrimination of both the inks using Raman and transmission IR spectroscopy.	Optimal Raman analysis of iron gall inks requires longer wavelengths (852 nm), while logwood inks show varying spectral responses with different excitation lines. This guides the choice of excitation line for analysing these ink types.	2024	[147]
Ink	Analysis of age of ink by Raman Spectroscopy and GC-MS.	Raman instrument has successfully determined the age of the inks by demonstrating the peak intensity of carbon black in signature samples.	2024	[148]
Water Colour ink	Identification of watercolour ink using Raman and FT-IR.	The powerful combination of FT-IR and Raman, coupled with advanced non-linear classification models, unlocks the secrets of water colour inks, enabling art conservators and researchers to precisely identify and distinguish between various ink types, and facilitating accurate restoration and authentication.	2024	[149]

statistical tools for analysing the organic constituents of GSR. As most of the organic residue is generated from the propellant of the ammunition, the reproducibility of this method was further studied in order to check its validity for routine analysis, which, on analysis, turned out to be a specific and potent approach for screening purposes[9].

Particles resembling GSR were also found in several other automobile products, such as air brakes, car brake pads, etc. In a recent study, scientists have proposed another method utilising luminescent markers for the identification of GSR from non-toxic ammunition. It allowed the reliable on-site detection of GSR [158]. Another approach was made, and a two-step methodology was developed to analyse organic gunshot residues from different ammunition. The method included the detection of fluorescent particles, followed by further characterisation using Raman spectroscopy. However, it must still be tested for false positives and for use with lead-free ammunition [156]. Diphenylamine and ethyl centralite are the important constituents of GSR. They are used as stabilisers of popular propellants. Raman spectroscopies have significantly distinguished bullet calibres from the analysis of chemical constituents collected 20 cm from the target [154]. Thayer et al. [159] applied SERS for the portable detection of these organic propellants in various solvents to maintain homeland security and forensic science.

Raman spectroscopy has proven to be a valuable tool for analysing GSR, enabling the detection of various components such as Mercury Chloride, Ethyl Centralite, and Diphenylamine [153]. Combined with chemometric analysis, Raman spectroscopy can discriminate GSR particles and their origin, providing indispensable insights [160]. This technique can also perform elemental analysis of organic and inorganic constituents, making it a versatile tool for GSR analysis. The technique has been successfully applied to smokeless powder analysis, allowing for the determination of the manufacturer and calibre of ammunition [161]. Furthermore, it enables non-destructive identification and differentiation of organic gunshot residue, with SVMDA modelling achieving excellent classification results [162]. Portable Raman spectroscopy and SERS techniques have also been shown to enhance detection accuracy and limits, facilitating the analysis of organic GSR.

The use of Raman spectroscopy in GSR analysis has far-reaching implications, including the ability to interpret elemental composition and molecular signatures of organic GSR. This technique has the potential to revolutionise the field of forensic science, providing a powerful

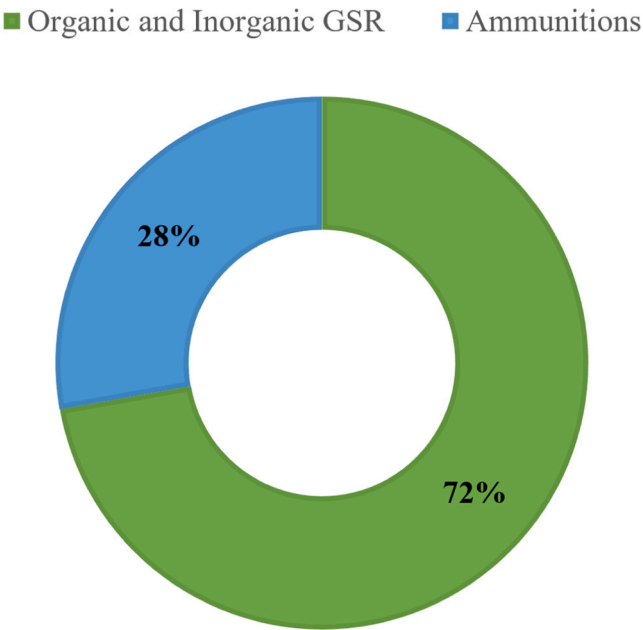


Fig. 7. Percentage distribution of Publications (2007-June2024) specific to GSR and ammunition analysis viz. Raman Spectroscopy.

tool for investigators to analyse GSR and other ballistic trace materials. Fig. 7 presents the distribution of research publications from 2007 June 2024 that analyse GSR, gunpowder, ammunition, and related materials using Raman spectroscopy."

Similarly, the significant studies and analytical work represented in the literature elaborating the analysis of ballistics trace materials using Raman spectrometer are incorporated in Table 3.

6. Analysis of paint traces

Paint is one of the most important categories of forensic traces found at various places; however, it finds greater relevancy in forensic cases, especially in vehicular accidents. The fragments of paint chips, or smears can be found at the scene of the occurrence, attached to the surroundings with which it came in contact, such as vehicles, roads, trees, hoardings in the case of accident and in walls, doors, furniture in the case of burglary, robbery etc. Like other traces paint chips can be exchanged on contact [171]. Paint is a characteristic marker of various vehicles; different vehicles have different coatings of paint, and the composition is a unique characteristic. Paint is the mixture of pigments, oils, driers, solvents, and other additives [172] as shown in Fig. 8.

Automotive or automobile paints and architectural paints are two major types of paint commonly encountered in hit-and-run cases [173]. Identification and comparison of the unknown paint samples to determine the origin of the paint are the primary goal of forensic examination of paint. Comparison of paint can be done primarily by physical/ visual matching and by microscopic examination by stereomicroscope. A unique database has already been created for the comparison of paint samples. Various forensic science laboratories across the world use Paint Data Query. Hence, instrumental analysis of paint is the major approach commonly used by various groups of forensic experts. This instrumental analysis of paint samples usually takes place with the help of FTIR, Pyrolysis GS-MS and SEM-EDX [171]. Other techniques, such as X-ray diffraction, NAA can be used to analyse the elemental composition of paint samples. Traces of automotive paints are typical in the cases of vehicular accidents.

Automotive finishes consist of primer, electrocoat, clear coat and base coat. These layers comprise polymers, organic, inorganic compound, resins etc. [174] μ -Raman spectroscopy and FTIR were used to analyse automotive paint for forensic purposes. In a study, a total of 10 automotive paint samples from topcoats was examined. Various organic and inorganic components were identified based on Raman spectra. However, the interferences from fluorescence were observed and hindered the identification of pigments [175]. IR and Raman spectroscopy can be utilized for routine forensic analysis of paint for identification and comparison. The Raman spectra of the sample obtained from the victim can be matched with the suspected sample. Using excitation lasers of different wavelength reduced the interference associated with fluorescence [176]. The interpretation of data obtained by spectroscopic techniques was usually accompanied with chemometric analysis. It reduces the chances of errors and overall analysis time [177]. The discrimination of automotive and household paint can also be achieved by this method [178]. In a similar study regarding the paint, a total of 348 samples were examined under Raman micro-spectroscopy. Of the total samples, 48 samples belonged to architectural paint and 300 samples were automotive paints. More than 498 layers were analysed and various pigments, extenders, additives, stabilisers and other components were discriminated [179]. In 2014, experts explored the applications of Raman microspectroscopy for the discrimination of solid and metallic paint samples. More than 90 % samples were characterised successfully, the majorly occurring problem with interpretation of spectra for comparison was still intact with the analysis [180]. Further, another method has been applied- a ratio-based approach- which helped obtain the Raman spectra of blue automotive paint sample of various cars, and the issues related to their comparison were resolved using databases of paints i.e., spectral databases.

Table 3

Available literature on the Forensic analysis of Gun-shot residue using Raman spectroscopy.

Sample	Aim	Results	Year	Ref.
Organic GSR	Detection of Organic GSR components using Carbon SEM Stubs hyphenated with Raman Spectroscopy	The detection of OGSR retained the ability of SEM-EDS for analysis	2010	[163]
Ammunition	Raman Spectroscopic Analysis for Gunshot Residue with New materials for Calibre Differentiation.	Raman spectra of tan-colored GSR particles after analysis have been proven as the most accurate method for identification of ammunition calibre–firearm pair	2012	[154]
Ammunition	Ammunition Identification by Means of the Organic Analysis of GSR Using Raman Spectroscopy	Raman spectroscopy has been proven to be a useful screening tool when a faster analysis of GSR traces is required	2012	[156]
GSR	Discrimination of GSR using Raman and FT-IR coupled with statistical tools	The two analytical methods provided complementary information	2013	[155]
Ammunition	The identification of inorganic and organic GSR particles of lead-free ammunitions were performed using laser ablation-ICPMS and Raman micro-spectroscopy	Raman micro-spectroscopy is a proven successful technique for the analysis of inorganic and organic GSR.	2014	[164]
Ethyl and methyl centralite	Analysis of ethyl (EC) and methyl centralite (MC) vibrational spectra for mapping OGSR.	Raman mapping of EC and MC is a high-throughput line-scan Raman microscopy.	2014	[165]
Antimony	Determination of Antimony by SERS	The study has highlighted a new method for antimony-based analysis using SERS spectra of an Antimony-phenylfluorone (SbPhF) complex.	2014	[166]
Ammunition	Raman Imaging is used for rapid analysis of GSR	The coupled Raman imaging and the MCR method have been used as a powerful tool for the organic analysis of the GSR particles on cloth samples.	2015	[167]
Gunpowder	SERS is used for the analysis of smokeless gunpowder and macroscopic GSR	SERS is highly specific and sensitive for GSR detection, and SEM-EDX is used significantly.	2016	[157]
Ammunition	Raman micro-spectroscopic mapping as a tool for the detection of	The method is highly specific and has differentiated samples obtained from	2018	[10]

(continued on next page)

Table 3 (continued)

Sample	Aim	Results	Year	Ref.
Organic GSR	gunshot residue on adhesive tape Raman and chemometric analyses were done for organic GSR obtained from different sources.	automotive mechanics and GSR particles. The study is potentially the most powerful tool for the identification and classification of Organic GSR.	2019	[168]
Luminescent markers of GSR	Determination of Luminescent Markers of GSR using Fluorescence, Raman Spectroscopy, with attached Chemometrics	Analysis of luminescent markers with reliable GSR from Non-Toxic Ammunition residues.	2019	[158]
Organic gunshot residue	A Novel Two-Step Method Used for Organic Detection of GSR	The analysis helped characterise of the fluorescent particles of organic GSR.	2019	[169]
Diphenylamine and Ethyl centralite	Detection of GSR	The signal detection limit (SDL), limit of detection (LOD), and limit of quantitation (LOQ) of DPA and EC in acetone, acetonitrile, ethanol, and methanol were measured.	2019	[159]
Organic GSR	Component analysis of GSR using Raman Spectrometry	Mercury Chloride, Ethyl Centralite, Diphenylamine, N-Nitrosodiphenylamine, 2-Nitrodiphenylamine, 4-Nitrodiphenylamine, 2,4-Dinitrodiphenylamine, 2,4-Dinitrotoluene, Barium Nitrate, Lead Sulfate, Antimony Sulfide has been successfully detected by Raman Spectroscopy.	2020	[153]
Gun Shot Residue	Use of vibrational spectroscopy with chemometric tools for analysis of GSR.	Raman Spectroscopy has been successfully able to discriminate GSR particles and their origin, and coupled chemometric analysis has been indispensable for this purpose.	2021	[160]
Inorganic and Organic GSR	Analysis of constituents in GSR	Raman Spectroscopy can be used for elemental analysis of both organic and inorganic constituents. Various components, such as Diphenylamine (DPA), Nitrocellulose (NC), N-Nitrosodiphenylamine (NNDPA), 2-Nitrodiphenylamine (2-NDPA), 4-Nitrodiphenylamine (4-NDPA), and Ethyl Centralite (EC), are detected in gunshot residue.	2021	[153]
Organic Gun Shot Residue	Determination of manufacturer and calibre of ammunition through smokeless powder analysis	Raman Spectroscopy has successfully analysed smokeless powder recovered from cartridge cases.	2022	[161]

Table 3 (continued)

Sample	Aim	Results	Year	Ref.
Organic GSR	Interpretation of elemental composition and molecular signatures of OGR using Raman Spectroscopy.	Raman spectroscopy and LIBS enable non-destructive identification and differentiation of organic gunshot residue, with SVMDA modelling achieving excellent classification results.	2023	[162]
Organic GSR	Analysis of Organic GSR facilitated by portable Raman Spectrometer and SERS.	Portable Raman and SERS techniques enhance the detection of organic gunshot residue, with computational analysis and pre-processing improving accuracy and limits.	2024	[170]

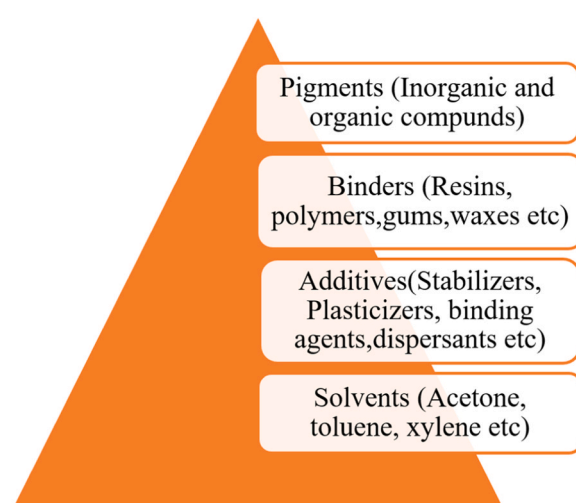


Fig. 8. Basic Components of Paint.

FTIR and Raman spectroscopy are complementary techniques for discriminating different paint samples. The primary advantage of Raman spectroscopy over FTIR is its ability to differentiate paint samples of the same colour [181]. FTIR effectively analyses additives and organic pigments present in paint, whereas Raman spectroscopy is able to identify all pigments and discriminate paint samples of red, white, yellow, green and blue colour. It is a rapid and accurate approach for identifying vehicles in hit-and-run cases for forensic investigation. [182]. The deterioration of paint on a vehicle is an observed phenomenon in an aged car. It is important from a forensic point of view, as analysis of various dependent factors is more significant. Oliveira [183] explored these factors and studied the effects of degradation in the composition of multi-layered paint by Raman spectroscopy to identify the vehicle from the database accurately. In further analysis, it was reported that the combination of Raman spectroscopy, ATR-FTIR and SEM-EDX was used to explore the heterogeneity of paint chips and to study and enhance the detailed changes occurring in the composition of the paint. Different techniques combined with Raman spectroscopy are being applied to analyse paint. The data can be analysed with the help of advanced chemometric approaches for forensic purposes [184]. The percentage distribution of research work published for the analysis of different types of paints including automotive paints, architectural, artistic, household and other types of paints using Raman Spectroscopy in several literature across 2007 - June 2024 is represented in Fig. 9.

In a recent experiment, paint chip samples from over 30 cars were taken and examined using Raman spectrometer. Variations in the

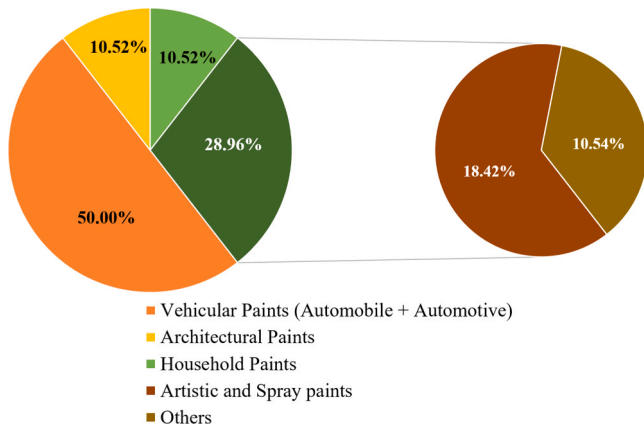


Fig. 9. Percentage distribution of Publications (2007-June 2024) specific to different types of paint analysis viz. Raman Spectroscopy.

composition of paint were seen at different places, even when the paint belonged to the same car[185]. The study of these changes provides useful information for forensic relevance. The same multimodal approach including Raman spectroscopy, FTIR and SEM-EDX, earlier tested for the detailed analysis of paint composition, is currently being tested for the characterization of multilayer paint conducted for the characterization of multilayered vehicular paints in a recent study. Various available literature sources (2007-June 2024) on the analysis of paint and pigment using Raman spectroscopy are depicted in Table 4.

7. Soil

Soil Pedology deals with the study of soil, including its physical characteristics, composition, and layers. Using pedological principles and knowledge of soil, forensic scientists examine disturbed soils which are frequently affected by human activity. By comparing the dirt materials found on tyres, shoes, clothing, and equipment from moving vehicles, this comparison helps connect suspects to crime scenes. The framework for analysing soil morphology is provided by pedology, which enables forensic specialists to make significant inferences from soil data. The accuracy of forensic investigations is improved by the capacity to examine not only the soil itself but also foreign objects like paint chips, broken glass, pollen, and microorganisms embedded in the soil. This type of trace material has proved to be helpful in several cases, such as hit-and-runs, kidnappings, sexual assaults, and crimes involving wildlife etc. Fitzpatrick et al. [210] highlighted the requirement of a systemic approach towards the crime scene in investigating soil samples from shoes, vehicles and crime scenes, demonstrating the importance of soil in a hit-and-run case.

The physical methods that forensic soil scientists use today include visual examination, colour and texture comparison, mineral identification, particle size distribution, and density gradient measurement. Organic matter is analysed using by techniques such as FTIR, nuclear magnetic resonance spectroscopy (NMR) and thermogravimetric analysis (TGA) based on thermal properties of soil. Energy-dispersive X-ray spectroscopy (EDX) and transmission electron microscopy (TEM) identify of inorganic and mineral particles. Elemental analysis techniques like X-ray spectroscopy, atomic absorption spectroscopy (AAS), neutron activation analysis (NAA), inductively coupled plasma mass spectrometry (ICP-MS), and optical emission spectrometry (ICP-OES). Methods such as scanning electron microscope (SEM), transmission electron microscope (TEM) are used to examine the morphology and chemical structure of soil. Raman spectroscopy can also provide chemical structure, phase and polymorph, crystallinity and molecular interactions [211]. Raman spectroscopy offers advantages as it requires minimal sample preparation and less cost unlike SEM/EDX [212]. For the detection of organic substances and minerals, a portable Raman

Table 4

Available literature review on the forensic analysis of paint using Raman spectroscopy.

Sample	Aim	Results	Year	Ref.
Automotive paints	The interpretation of paint using population studies	The discrimination among different was obtained from Raman spectroscopy.	2007	[186]
Architectural	The interpretation of various paint traces	Raman spectroscopy allowed the detection of different colour from traces.	2007	[186]
Automotive	Automotive Topcoats analysis	The pigments used in those automotive topcoats have been identified by their Raman spectra.	2008	[187]
Automobile paint	Examination of forensic automotive paint samples	The coupling of Raman and X-ray fluorescence (XRF) allowed forensic experts to examine paint samples using a non-destructive paint analysis.	2008	[102]
Painting/ artistic paints	Spectroscopic investigation of external wall paintings from St. Dumitru's Church, Suceava, Romania	External wall paintings have been analysed using the micro-Raman spectroscopy.	2008	[188]
Household Paint	Forensic analysis of household paint samples	The IR spectra are sensitive to binders and extenders, and the Raman spectra are sensitive to organic pigments.	2011	[177]
Automotive	Characterisation of paint samples by IR and Raman spectroscopy.	Both IR and Raman spectroscopy are used to examine of paint samples.	2011	[176]
Automotive and household paint	An investigation into the use of micro-Raman spectroscopy for the analysis of car paints and single textile fibres	Micro-Raman spectroscopy is a good tool for forensic identification of pigments present in micro paint chips and dyes.	2011	[175]
Architectural paint	Forensic examination of multilayer white paint by lateral scanning Raman spectroscopy	Cross-sections of even optically indistinguishable white paint layers readily reveal the composition of the different layers of paints.	2012	[189]
Automotive Paint	Characterization of Blue Pigments Used in Automotive Paints by Raman Spectroscopy	Raman spectra provide information about the primary pigments present in the sample.	2012	[180]
Automotive paint	Application of IR and Raman Spectroscopy in examination of paint traces.	IR and Raman are the best analytical tools for forensic paint analysis.	2013	[190]
Architectural and automotive paint	Raman Spectroscopy of Automotive and Architectural Paints	Discrimination of pigments within paint samples can be done using Raman spectroscopy.	2013	[179]
Automotive	Interpretation of FTIR spectra of polymers and Raman spectra of car paints by wavelet transform for	In combination with the chemometrics, successful analysis was done	2014	[191]

(continued on next page)

Table 4 (continued)

Sample	Aim	Results	Year	Ref.
Spray paints	reducing data dimensionality Chemical characterisation of spray paints by a multi-analytical (Py/GC-MS, FTIR, μ -Raman) approach	Micro-Raman spectroscopy and microscopy have discriminated inorganic pigment, extenders, azo pigments and some polycyclic pigments. Raman spectroscopy increases the discrimination efficiency of the samples and is also non-destructive.	2016	[192]
Automotive paint	Analysis of 52 automotive coating samples for forensic purposes with FTIR and Raman microscopy	This is an effective tool in identification of paints pigments in hit-and-run cases with confocal Raman microscope	2016	[182]
Automotive paint	Effective identification of paints pigments in hit-and-run cases with confocal Raman microscope	Multiblock tools, especially CCSWA (common component and specific weight analysis), have been used for the demonstrated of origin of paint samples.	2016	[181]
Household paint	Combining spectroscopic data in the forensic analysis of paint utilises a multiblock chemometric tool	Forensic scientists must have huge excitation, especially Near IR 785 nm.	2017	[37]
Automotive paint	Raman spectroscopy of automotive paints: Forensic analysis of variability and spectral quality	Raman spectroscopy imaging paint chips more effectively evaluate the degradation processes and molecular composition.	2018	[183]
Automotive paint	IR and Raman spectroscopy were used for automotive paints analysis	Raman paint spectra on aluminium and glass sheet enable the identification of the kind of paint, while spectra of paint deposited on fabrics are more complicated.	2018	[193]
Spray Paint	The influence of the type of substrate on the possibility of spray paint identification for forensic purposes	A combined Raman-SERS is non time dependent tool and is used for standardizing Raman measurements.	2018	[194]
Headlight dyes/automotive	Contribution of Raman and SERS to the analysis of vehicle headlights: dye(s) characterization	Data from spectra uses the far-IR region (50 cm^{-1}), a valuable tool to ATR-FTIR for determination of many species.	2019	[195]
Automotive paint	Multi-Modal analysis by ATR-FTIR Imaging, Raman Microspectrometry, and SEM/EDX coupling for paint analysis	Raman spectroscopy distinguishes paint production batches and identifies recycled paints with molecular fingerprints. Combined with other techniques, it	2019	[196]

Table 4 (continued)

Sample	Aim	Results	Year	Ref.
Automotive paint	Analysis of the composition and layers of vehicular paint by coupled FT-IR and Raman spectroscopies	confirms findings with manufacturer recipes and raw materials. The analysis is done with Raman spectroscopy, which has differentiated dyes based on organic composition. Fourier-transform Raman spectroscopy and SERS identified primary paint dyes. Thin-layer chromatography with SERS distinguished additional chromophore mixtures. This methodology informs art restoration and retouching efforts.	2020	[185]
Fluorescent artists paints	Analysis of fluorescent paints using simple FT-IR, Raman and TLC-SERS	Raman and FT-IR have relentlessly distinguished paint binder's structural and chemical composition of to distinguish pain deposited on different kind of individual vehicles. The study is potent for providing actual results for determining of architectural paints.	2020	[197]
Vehicular/Automotive paint chips	Raman and FT-IR has been used for analysis of paint samples from different regions within a car.	FT-IR and FT-Raman-based characterisation has been proven to be most evident techniques for the analysis of two quinacridone pigments (PR122 & PV19) in artists' paint samples. A handheld device successfully identified key components in varnished oil paints, simplifying the analysis process and making it more accessible. This breakthrough enables easier art conservation and restoration.	2021	[185]
Architectural paint	White architectural paints were distinguished by microscopic by Raman spectroscopy and associated chemometrics	It has been evident that Raman is a better tool for analysing of automotive clear coats and paint smears when coupled with chemometrics, than IR microscopy.	2021	[198]
Contemporary artists' paints	Identification of organic synthetic pigments in artists' paints using FT-IR and FT-Raman.		2021	[199]
Ancient Synthetic organic paints	Analysis of synthetic organic pigments in pre-1950 paint samples using Handheld Raman.		2021	[200]
Automotive clear coats and paint smears	Raman and IR Microscopic analysis to examine forensic paint samples automotive clear coats and paint smears.		2022	[201]

(continued on next page)

Table 4 (continued)

Sample	Aim	Results	Year	Ref.
Artistic/ Canvas paint	Raman spectroscopic analysis for analysis of Canvas painting	Raman spectroscopy is one of the most powerful tools for analysing of micro characteristics of canvas paintings.	2022	[202]
Multilayer Automotive Paint	Analysis of cross-sectional area of multiple layers of automotive paints using Raman and LIBS.	An important tool in determining inherent complexity of multiple layers of automotive paints for origin identification. Multivariate data analysis and Raman successfully investigated the molecular and elemental characterisation of automotive paints.	2022	[203]
Multilayered automotive coatings	This study uses FTIR, Raman spectroscopy, and SEM/EDX to identify and distinguish different automotive coatings for forensic purposes.	These combined techniques enable accurate analysis and comparison of coating samples, aiding in investigations and trace analysis.	2022	[204]
Artists' Spray paints	Chemical profiling of spray paints using instrumental techniques such as Raman, ATR-FTIR, SEM/EDS, etc.	Raman spectroscopy detects binders, pigments & additives in over paintings, identifying CuPc polymorphs, alongside ATR-FTIR and SEM/EDS for accurate artwork conservation.	2023	[205]
Spray paints	FTIR and Raman spectroscopy coupled with chemometrics for determination of chemical profile of paints.	FTIR and Raman has estimated it to be most indispensable tool for determining the ageing and discrimination of the weathering process of paints.	2023	[206]
White Automotive Paint	White automotive paint was analysed with Raman Spectroscopy coupled with chemometric for identification of manufacturer.	Raman Spectroscopy can provide good discriminant power for analysing different layers of paints.	2024	[207]
Lipstick samples	Discrimination of different lipstick samples using FT-IR and Raman spectroscopy,	Vacuum FT-IR and Raman coupled PCA successfully differentiated 20 pink shade lipsticks from the same manufacturer with a 95.8 % accuracy rate.	2024	[208]
Parchment medieval paints	Characterisation and analysis of medieval paints on parchment (animal skin) using Handheld Raman and micro-SORS.	Combining Raman and micro-SORS has been conventional in the analysis of multiple components present in these paints.	2024	[209]

spectrometer is considered an excellent choice for outdoor fieldwork [213]. The advantages of providing results at the lowest temperature of -25°C and its portability make it a considerable choice for soil examination for forensic purposes [214]. Confocal Renishaw can identify astrobiological components such as crystalline calcite and gypsum via a Raman microscope at 785 nm. Its limitations include the requirement for higher spectral resolution and the potential for fluorescence interference [215].

Oxidative pre-treatment of soil samples with hydrogen peroxide can be used to overcome this issue. Hydrogen peroxide is usually used to remove aliphatic and aromatic contaminants from the soil. With increasing concentrations of hydrogen peroxide, significant improvement has been observed in fluorescence levels and the noise-to-signal ratio of the spectra [216]. Additionally, applying acid digestion with oxidative treatment helped improve the background interference, allowing compounds to be identified more clearly. Acid digestion with concentrated sulphuric acid and nitric acid removes the organic component broken down by oxidative treatment, leaving the rest inorganic material for analysis [217].

Gypsum, brushite, cerussite and calcite are not primary components of most soil, but their presence can be valuable in various investigations. Raman spectroscopy can potentially differentiate these components, which may appear similar in colour, preventing misidentification in a burial context [218].

Raman Spectroscopy primarily focuses on identifying the chemical composition of mineral material. It cannot determine the particle size and shape required for the classification of soil. Morphologically directed Raman Spectroscopy offers an advantage over Raman spectroscopy, combining automatic particle imaging and allowing the determination of physical and chemical characteristics of soil with better accuracy and sensitivity [219].

Confocal micro-Raman spectroscopy was also tested to detect species of calcium phosphate commonly found in soil. Despite fluorescence interference, the technique could differentiate all three species of calcium phosphate [220].

SOM stands for soil organic matter, which is composed of organic material, including decomposed plants, animals, humus, and microbes. From a forensic perspective, it can help classify the soil found at the crime scene, which may help link the soil to a specific location or place. A study on soil found that Raman Spectroscopy and Attenuated total reflectance (ATR- FTIR) have the potential to be used for analysing SOM, but with limitations. ATR FT-IR focuses on determining functional groups in the organic matter, whereas Raman spectroscopy identifies the molecular structure of organic and inorganic matter. They are complementary techniques and can be used together. By combining these two techniques, many limitations can be resolved.

The accuracy of the combination in forensic soil analysis can be enhanced by introducing advanced chemometric method, one of which is Competitive Adaptive Reweighted Sampling commonly known as CARS. It reduces noise by selecting the most suitable variables, ultimately improving the predictive model [221]. Not just with ATR FT-IR, the fusion of Raman spectroscopy and near-infrared spectroscopy can also be utilized to analyze SOM, along with modelling algorithms to enhance the accuracy [222]. The problem of fluorescence may still arise in some soils, such as sandy soil, which is primarily composed of fine quartz particles. These interferences present obstacles in characterizing soil using Raman Spectroscopy [223]. One solution to such analysis can be SERS or different excitation lasers other than commonly used 785 nm, which should be explored for quartz-based fine particles in sandy soil found in arid environments [224]. Various available literature (2007-June 2024) on analysis of soil using Raman spectroscopy has been depicted in Table 5.

Raman Spectroscopy has been extensively applied in soil analysis, covering various aspects of soil science, environmental monitoring, and forensic investigations. Research topics include the detection of uranium oxides, soil humus and humic acids, investigation of iron corrosion,

Table 5

Available literature review on the forensic analysis of soil using Raman spectroscopy.

Sample	Aim	Results	Year	Ref.
Soil	Determination of uranium oxides in particulate matter in soil samples extracted from uranium mines, with the help of μ -Raman spectrometry	Uraninite (UO ₂) residues and alteration products were detected, displaying distinctive Raman spectral features due to their sensitivity to laser wavelength.	2008	[225]
Soil	The simultaneous application of Surface-Enhanced Raman Scattering (SERS) and Surface-Enhanced Fluorescence (SEF) spectroscopies enabled the comprehensive analysis of soil humus and humic acids.	This provides valuable insights into their molecular composition and structural properties. The study investigated pH, concentration, and amendment effects on their structure.	2008	[226]
Anoxic soils	Investigate long-term corrosion mechanisms of iron in anoxic soils using archaeological artefacts	The study provides insights into iron corrosion mechanisms in anoxic soils, which are valuable for cultural heritage, oil and gas, and nuclear waste disposal contexts. It also helps identify siderite, chukanovite, and magnetite in corrosion layers.	2010	[227]
Soil	To analyse mineralogical, organic, and water-soluble fractions in soil samples using Raman Spectroscopy.	Successful discrimination between urban and rural soil types based on mineralogy and organic/water-soluble fractions.	2011	[216]
Lunar Soils	Investigate mineralogy of < 45 μ m grains in four lunar soils using Laser Raman spectroscopy.	Identified major and minor lunar minerals through Raman spectral features. Calculated compositional distributions of pyroxene and olivine.	2011	[228]
Soil and herbicides	Investigate interaction between herbicide paraquat and soil humic acids using surface-enhanced Raman scattering (SERS) and fluorescence spectroscopy.	Successful detection of paraquat-humic acid complex formation, influenced by pH and laser excitation wavelength.	2011	[229]
Soil	Characterize diadochite and destinezite minerals in soils using Raman spectroscopy.	Distinct Raman spectra for amorphous diadochite and crystalline destinezite, specific bands to phosphate, sulfate, and water stretching/bending modes were assigned.	2011	[230]
Soil	Review spectroscopic techniques for analysing phosphorus species in environmental matrices (soil).	Four techniques (XANES NMR, FTIR, Raman) identified common P species in soils, sediments, and waste as	2011	[231]

Table 5 (continued)

Sample	Aim	Results	Year	Ref.
		Hydroxyapatite and octa calcium phosphate minerals, and Phosphate adsorbed on Fe- and Al-oxides		
Soil	Analysis of soil phosphorus content in using Raman Spectroscopy	Prediction of soil phosphorus concentration using PLS modelling after wavelet packet signal extraction. Calibration R ² reached 1 and validation R ² was 0.937, demonstrating high potential for soil phosphorus detection.	2012	[232]
Roadside soils and plants	Investigate impact of traffic on roadside soils and plants using Raman spectroscopy.	Analysis revealed guardrails as a significant source of Zn contamination in underlying soils, explaining anomalous Zn concentrations detected in previous studies.	2012	[233]
Indian Dark Earths Soil	To characterize the stable carbon content in Terra Preta de Índio (Indian Dark Earths) soils using Raman spectroscopy.	Raman analysis identified sp ² phase dominance, surface/core disorder variations, and distinguished TPI-carbon from charcoal.	2012	[234]
Urban soils	Investigate synergistic impacts of guardrails and traffic pollutants on urban soils using Raman spectroscopy, XRF, and chemical simulation.	The study reveals significant contribution of guardrails to urban soil pollution through zinc release and highlights potential environmental risks, emphasizing need for mitigation strategies.	2012	[235]
Soil Gases	Develop an online method for quantifying soil gases using combined FTIR and Raman spectrometry for CO ₂ storage site monitoring.	Successful continuous measurement of N ₂ , O ₂ , H ₂ O, and CO ₂ concentrations in soil using a coupled Raman/FTIR system.	2013	[236]
Soil	Reviewing vibrational spectroscopy techniques (FTIR & Raman) for soil chemistry research	Comprehensive overview of applications in soil chemistry, highlighting versatility and complementarity of FTIR and Raman spectroscopy for analysing environmental samples.	2014	[237]
Soil	Develop a rapid soil classification method using Raman spectroscopy combined with chemometric tools.	A robust classification method using Raman spectroscopy, Support Vector Machines (SVM-C), and Genetic Algorithm (GA) achieved 100 % accuracy, validated by X-ray fluorescence spectrometry.	2014	[238]
Desert Soil	Test the Mars Micro-beam Raman Spectrometer (MMRS) on a rover in	Successful identification of igneous minerals, carbonates, sulfates,	2015	[239]

(continued on next page)

Table 5 (continued)

Sample	Aim	Results	Year	Ref.
	the Atacama Desert to simulate Mars surface exploration.	and carbonaceous materials. Quantified distributions of major minerals and carbonaceous materials.		
Chinese farmland Soil	Raman spectroscopy's potential was explored for soil identification and prediction of SOM.	200 Chinese farmland soil samples were analysed using Raman spectroscopy (180–3200 cm ⁻¹), with PLS for SOM prediction and Probabilistic Neural Network (PNN) with PCA for soil identification.	2016	[240]
Japanese soil samples	Analysis of Japanese soil samples using Raman Spectroscopy	Raman spectroscopy analysed charcoal fragments in Japanese soils, comparing ancient (1000-year-old) and modern samples. Results showed Charcoal fragments alter chemically over decades/centuries, indicated by graphitization changes.	2017	[241]
Different soil samples	To evaluate Morphologically Directed Raman Spectroscopy (MDRS) for soil analysis and distinguishing between soils, in forensic investigations.	MDRS analysis revealed significant geographical differences in soil mineral content, identifying distinct variations between and within sites, proving its reliability for forensic analysis.	2018	[219]
Soil samples	Detection of Thiabendazole (TBZ) in Soil using Surface-Enhanced Raman Spectroscopy (SERS)	The study achieved precise TBZ detection in soil via SERS and AuNPs, with a Limit of Detection (LOD) of 0.1 mg/L, Limit of Quantification (LOQ) of 1 mg/L, accuracy (Rp2) of 0.9769, and error (RMSEP) of 0.556 mg/L, and recovery ranging from 95.90 % to 116.65 %.	2018	[242]
Soil samples	Gold Nanoparticles for Qualitative Detection of Deltamethrin and Carbofuran Residues in Soil by SERS	AuNPs (Gold-nanoparticles)-based SERS technique provides accurate detection of pesticide residues in soil, offering technical support for environmental monitoring	2019	[243]
Soil samples	Shifted Excitation Raman Difference Spectroscopy (SERDS) for Soil Analysis	SERDS proves to be a promising technique for soil analysis, overcoming background signal limitations and enabling accurate detection of mineral components.	2021	[244]
Soil samples	Combining FTIR-ATR and Raman	Combining FTIR-ATR and Raman	2021	[221]

Table 5 (continued)

Sample	Aim	Results	Year	Ref.
	Spectroscopy for prediction of Organic matter in soil samples	spectroscopy with CARS-based data fusion provides an accurate and rapid method for predicting soil organic matter, offering a promising approach for soil fertility assessment.		
Soil and river samples	Detection of Benzo [a]pyrene (BaP) using Surface-Enhanced Raman Scattering (SERS)	The AgNR substrate enables sensitive SERS detection of BaP in environmental samples, showing promise for monitoring carcinogenic pollutants.	2021	[245]
Different soil samples	Raman spectroscopic analysis for chemical composition of soil.	Spectroscopic techniques offer a promising approach for soil analysis, providing valuable information on chemical composition. Further research is needed to address existing gaps and optimize these methods for various soil science applications.	2021	[246]
Agricultural soils	Portable Shifted Excitation Raman Difference Spectroscopy (SERDS) Sensor for On-Site Soil Analysis in Precision Agriculture	The portable SERDS sensor demonstrates high potential for on-site soil analysis in precision agriculture, offering rapid and accurate measurements for optimized crop management.	2022	[247]
Soil samples	Soil Fertility Analysis using Shifted Excitation Raman Difference Spectroscopy (SERDS)	Recovered Raman signals from inorganic (quartz, feldspar, anatase, calcite) and organic (amorphous carbon) constituents. Predicted soil organic matter content ($R^2 = 0.82$, Root Mean Squared Error of Cross-Validation = 0.41 %)	2022	[248]
Agricultural Soil	Detection of calcium phosphate species in soil by confocal μ -Raman spectroscopy	Confocal spectroscopy helps detect phosphorus species (inorganic compound) in soil.	2022	[220]
Soil	Negative result: Application of Raman spectroscopy to the forensic analysis of an arid, sandy, soil	Raman spectroscopy is found to be unsatisfactory technique for the analysis of fine fraction quartz recovered from the Swan Coastal Plain soil.	2023	[224]
Soil	Improving the estimation accuracy of soil organic matter based on the fusion of near-infrared and RS using the outer-product analysis	Combining Raman Spectroscopy with infrared spectroscopy improves accuracy for the estimating soil organic matter.	2024	[222]

mineralogical analysis, and herbicide interaction with soil humic acids. Techniques such as micro-Raman spectrometry (μ -Raman), SERS, and Shifted Excitation Raman Difference Spectroscopy (SERDS) have been employed. The goals include soil classification and identification, forensic soil analysis, detection of pollutants and contaminants, soil fertility analysis, and environmental monitoring. Applications extend to precision agriculture, planetary exploration, and the simulation of Mars surface exploration. Overall, Raman spectroscopy has demonstrated versatility and potential in advancing our understanding of soil chemistry and addressing environmental challenges.

7.1. Raman spectroscopy, chemometrics and artificial intelligence

The Raman spectrum provides the molecular fingerprint of a compound, and with the help of chemometrics tools, complex spectral data can be extracted, effectively identifying important patterns and relationships. With the development of artificial intelligence, chemometrics tools are increasingly being integrated with machine learning to improve both accuracy and overall analytical performance [249]. Chen et al. [250] in their study on the chicken samples using SERS, applied particle swarm optimization- widely used in machine learning, to the SVM model, resulting in improved classification accuracy.

Dai et al. [251] integrated laser-induced breakdown spectroscopy with Raman spectroscopy and enhanced the accuracy of mineral identification using machine learning models such as PLS-DA, for geological purposes.

Xue et al. [252] applied the properties of Convolutional Neural Network (CNN) in combination with Raman spectroscopy to identify a species of fungal pathogen *Candida auris*, in clinical settings. They found the detection method achieved an accuracy of 93.33 % in identifying the species individually. Artificial Intelligence significantly enhances the accuracy and reduces the requirement for laboratory culturing or labelling.

There are also studies on the use of CNNs for the classification and interpretation of Raman spectra from biological specimens in the medical field, particularly for diagnosing diseases like hepatitis B and cancer. Biological fluids such as urine, saliva and semen play an important role in reconstructing crime scenes and identifying the culprit. However, the presence of body fluid on different substrates causes spectral interference, which complicates the interpretation of spectra. In cases involving mixtures of body fluids, these interferences further complicate the analysis, especially when using commonly applied regression techniques such as Least Squares (LS) and Principal Component Regression (PCR). These techniques often require extensive calibration for all possible substrates, limiting their applicability in forensic examinations [253]. In such cases, spectral decomposition-based techniques such as reducing spectrum complexity and Multivariate Curve Resolution offer advantages in the analysis of body fluid. However, some degree of errors remains due to variations caused by handling and environmental conditions. In their study of body fluids, Borisov et al. suggested using AI-based techniques such as PCA and deep learning to improve the accuracy of forensic analysis of biological fluid [254].

Field portable, rapid, and automated detection of commonly found ignitable liquids such as gasoline- through the integration of Raman spectroscopy with a neural network offers a significant advantage over traditional forensic techniques. The combined approach provides time efficiency and accurate, hands-free detection of the liquids [255]. Another application of CNN combined with Raman microscopy and chemometrics, is the classification of a commonly found alkaloid in the plant *Cannabis sativa*, which is a high significance in forensic chemistry and toxicology [256]. Techniques such as PCA, SVM and CNN have been shown in some studies to enhance the overall performance of detection and classification using Raman spectroscopy for various purposes [249]. Future research is expected to lead to significant advancements, helping to address current limitations

8. Conclusion

Raman spectroscopy is a cutting-edge technique used to analyse forensic traces owing to its non-destructive nature. The current review article on the analysis of forensic traces, such as inks, GSR, soil, and dyes, highlights the growing importance of Raman spectroscopy in forensic science. Despite its potential, Raman spectroscopy is currently employed in only a limited number of forensic laboratories in India. However, the use of portable handheld Raman spectrometers can simplify the initial screening of trace materials, thereby reducing the workload of forensic labs.

According to the available literature, India has made little to no contribution to the development or widespread adoption of this technique. Given the portability and efficiency of these instruments, there is a pressing need to adopt them for non-destructive and rapid analysis, including on-site crime scene investigations. Once implemented and integrated into routine casework, Raman spectroscopy has the potential to revolutionize forensic analysis. Nonetheless, its application in everyday forensic practice remains in its early stages and may take several years to be fully realized.

CRedit authorship contribution statement

Lovlish Gupta: Writing – review & editing, Methodology. **Samiksha Chauhan:** Writing – original draft, Methodology, Formal analysis. **sharma Sweety:** Writing – review & editing, Visualization, Supervision, Conceptualization.

Ethics approval

Not applicable.

Funding

Nil

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. Therefore, no external influences or biases have shaped the content of this paper.

Acknowledgement

The authors heartily acknowledge the administrative support of the National Forensic Sciences University, Delhi Campus for all the help and support and providing research conducive environment for framing this manuscript.

References

- [1] Crime Data: 2021 Nationwide Crime Statistics — LEB n.d. <https://leb.fbi.gov/bulletin-highlights/additional-highlights/crime-data-2021-nationwide-crime-statistics> (accessed January 31, 2023).
- [2] Criminal Investigation Process and the Role of Forensic Evidence: A Review of Empirical Findings | Office of Justice Programs n.d. <https://www.ojp.gov/ncjrs/virtual-library/abstracts/criminal-investigation-process-and-role-forensic-evidence-review> (accessed January 31, 2023).
- [3] C. Roux, R. Bucht, F. Crispino, P. De Forest, C. Lennard, P. Margot, et al., The Sydney declaration – revisiting the essence of forensic science through its fundamental principles, *Forensic Sci. Int* 332 (2022) 111182, <https://doi.org/10.1016/J.FORSCIINT.2022.111182>.
- [4] M.M. Evans, P.A. Stagner, Maintaining the chain of custody evidence handling in forensic cases, *AORN J.* 78 (2003) 563–569, [https://doi.org/10.1016/S0001-2092\(06\)60664-9](https://doi.org/10.1016/S0001-2092(06)60664-9).
- [5] M. Airlie, J. Robertson, M.N. Krosch, E. Brooks, Contemporary issues in forensic science—Worldwide survey results, *Forensic Sci. Int* 320 (2021) 110704, <https://doi.org/10.1016/J.FORSCIINT.2021.110704>.

- [6] M.B. Eyring, P. Martin, Spectroscopy in forensic science, Ref. Modul. Chem. Mol. Sci. Chem. Eng. (2013), <https://doi.org/10.1016/B978-0-12-409547-2.05455-X>.
- [7] (PDF) Ultra-Violet and Visible (UV–VIS) Spectroscopy and Chemometrics Techniques for Forensic Analysis of Ballpoint Pen Inks: A Preliminary Study n.d. (https://www.researchgate.net/publication/299445303_Ultra-Violet_and_Visible_UV-VIS_Spectroscopy_and_Chemometrics_Techniques_for_Forensic_Analysis_of_Ballpoint_Pen_Inks_A_Preliminary_Study) (accessed January 31, 2023).
- [8] C.S. Silva, A. Braz, M.F. Pimentel, Vibrational spectroscopy and chemometrics in forensic chemistry: critical review, current trends and challenges, J. Braz. Chem. Soc. 30 (2019) 2259–2290, <https://doi.org/10.21577/0103-5053.20190140>.
- [9] J. Bueno, I.K. Lednev, Raman microspectroscopic chemical mapping and chemometric classification for the identification of gunshot residue on adhesive tape, Anal. Bioanal. Chem. 406 (2014) 4595–4599, <https://doi.org/10.1007/S00216-014-7874-9>.
- [10] J. Bueno, L. Halámková, A. Rzhetskii, I.K. Lednev, Raman microspectroscopic mapping as a tool for detection of gunshot residue on adhesive tape, Anal. Bioanal. Chem. 410 (2018) 7295–7303, <https://doi.org/10.1007/S00216-018-1359-1>.
- [11] C.K. Muro, K.C. Doty, J. Bueno, L. Halámková, I.K. Lednev, Vibrational spectroscopy: recent developments to revolutionize forensic science, Anal. Chem. 87 (2015) 306–327, <https://doi.org/10.1021/AC504068A>.
- [12] H. Mitsutake, R.J. Poppi, M.C. Breitkreitz, Raman imaging spectroscopy: history, fundamentals and current scenario of the technique, J. Braz. Chem. Soc. 30 (2019) 2243–2258, <https://doi.org/10.21577/0103-5053.20190116>.
- [13] Using Raman Spectroscopy in Forensic Science n.d. <https://www.azom.com/article.aspx?ArticleID=11733> (accessed March 1, 2022).
- [14] E.M. Suzuki, P. Buzzing, Applications of Raman spectroscopy in forensic science. II: analysis considerations, spectral interpretation, and examination of evidence, Forensic Sci. Rev. (2018).
- [15] “Theory of Raman Spectroscopy” by David W. Ball n.d. (https://works.bepress.com/david_ball/115/) (accessed May 18, 2023).
- [16] G. Keresztury, Raman spectroscopy: theory, Handb. Vib. Spectrosc. (2001), <https://doi.org/10.1002/0470027320.S0109>.
- [17] R. Pilot, R. Signorini, C. Durante, L. Orian, M. Bhamidipati, L. Fabris, A review on Surface-Enhanced Raman scattering, Biosens. (Basel) 9 (2019), <https://doi.org/10.3390/BIOS9020057>.
- [18] (PDF) Raman Spectroscopy, a review n.d. (https://www.researchgate.net/publication/309179824_Raman_Spectroscopy_a_review) (accessed May 18, 2023).
- [19] H. Szymanski, Raman spectroscopy, Springer Science & Business Media, 2012, p. 256, <https://doi.org/10.1007/978-1-4684-3027-1>.
- [20] Smith E., Dent G. Modern Raman Spectroscopy. Modern Raman Spectroscopy 2019. <https://doi.org/10.1002/9781119440598>.
- [21] T.H. Kauffmann, N. Kokanyan, M.D. Fontana, Use of Stokes and anti-Stokes Raman scattering for new applications, J. Raman Spectrosc. 50 (2019) 418–424, <https://doi.org/10.1002/JRS.5523>.
- [22] F. Tian, Y. Zhang, J. Zhang, C. Pan, Raman spectroscopy: a new approach to measure the percentage of anatase TiO₂ exposed (001) facets, J. Phys. Chem. C. 116 (2012) 7515–7519, <https://doi.org/10.1021/JP301256H/ASSET/IMAGES/MEDIUM/JP-2012-01256H.0001.GIF>.
- [23] Vandenberghe P. Practical Raman Spectroscopy – An Introduction. Practical Raman Spectroscopy – An Introduction 2013. <https://doi.org/10.1002/9781119961284>.
- [24] S. Boyd, M.F. Bertino, S.J. Seashols, Raman spectroscopy of blood samples for forensic applications, Forensic Sci. Int. 208 (2011) 124–128, <https://doi.org/10.1016/J.FORSINT.2010.11.012>.
- [25] T. Casey, E. Mistek, L. Halámková, I.K. Lednev, Raman spectroscopy for forensic semen identification: method validation vs. Environmental interferences, Vib. Spectrosc. 109 (2020) 103065, <https://doi.org/10.1016/J.VIBSPEC.2020.103065>.
- [26] A. Sikirzhitskaya, V. Sikirzhitskiy, I.K. Lednev, Raman spectroscopic signature of vaginal fluid and its potential application in forensic body fluid identification, Forensic Sci. Int. 216 (2012) 44–48, <https://doi.org/10.1016/J.FORSINT.2011.08.015>.
- [27] K. Virkler, I.K. Lednev, Forensic body fluid identification: the Raman spectroscopic signature of saliva, Analyst 135 (2010) 512–517, <https://doi.org/10.1039/B919393F>.
- [28] S.Y. Ong, A. Wain, L. Groombridge, E. Grimes, Forensic identification of urine using the DMAC test: a method validation study, Sci. Justice 52 (2012) 90–95, <https://doi.org/10.1016/J.SCIJUS.2011.07.007>.
- [29] J. Filik, N. Stone, Analysis of human tear fluid by Raman spectroscopy, Anal. Chim. Acta 616 (2008) 177–184, <https://doi.org/10.1016/J.ACA.2008.04.036>.
- [30] M.J. West, M.J. Went, Detection of drugs of abuse by Raman spectroscopy, Drug Test. Anal. 3 (2011) 532–538, <https://doi.org/10.1002/DTA.217>.
- [31] E. Gerace, F. Seganti, C. Luciano, T. Lombardo, D. Di Corcia, H. Teifel, et al., On-site identification of psychoactive drugs by portable Raman spectroscopy during drug-checking service in electronic music events, Drug Alcohol Rev. 38 (2019) 50–56, <https://doi.org/10.1111/DAR.12887>.
- [32] F. Salahioglu, M.J. Went, S.J. Gibson, Application of Raman spectroscopy for the differentiation of lipstick traces, Anal. Methods 5 (2013) 5392–5401, <https://doi.org/10.1039/C3AY41274A>.
- [33] F. Salahioglu, M.J. Went, Differentiation of lipsticks by Raman spectroscopy, Undefined 223 (2012) 148–152, <https://doi.org/10.1016/J.FORSINT.2012.08.018>.
- [34] M.N.M. Asri, R. Verma, M.H. Ibrahim, V. Sharma, N.A.M. Nor, Rapid non-destructive techniques to identify the traces of kajal using chemometrics; a comparison of ATR-FTIR and Raman spectroscopy, Microchem. J. 169 (2021) 106556, <https://doi.org/10.1016/J.MICROC.2021.106556>.
- [35] M. López-López, J. Vaz, C. García-Ruiz, Confocal Raman spectroscopy for the analysis of nail polish evidence, Talanta 138 (2015) 155–162, <https://doi.org/10.1016/J.TALANTA.2015.02.031>.
- [36] H. Yang, J. Irudayaraj, M.M. Paradkar, Discriminant analysis of edible oils and fats by FTIR, FT-NIR and FT-Raman spectroscopy, Food Chem. 93 (2005) 25–32, <https://doi.org/10.1016/J.FOODCHEM.2004.08.039>.
- [37] K.B. Ferreira, A.G.G. Oliveira, J.A. Gomes, Raman spectroscopy of automotive paints: forensic analysis of variability and spectral quality, Spectrosc. Lett. 50 (2017) 102–110, <https://doi.org/10.1080/00387010.2017.1288635>.
- [38] E.R. Mojica, Z. Dai, New Raman spectroscopic methods' application in forensic science, Talanta Open 6 (2022) 100124, <https://doi.org/10.1016/J.TALO.2022.100124>.
- [39] A. Kazlagić, E. Omanović-Mikličanin, Application of Raman spectroscopy in food forensics: a review, IFMBE Proc. 73 (2020) 257–263, https://doi.org/10.1007/978-3-030-17971-7_40/COVER.
- [40] B. Thirumalraj, Surface enhanced Raman scattering spectroscopy (SERS) analysis of ink pens in forensic usage, Biomed. J. Sci. Tech. Res 16 (2019), <https://doi.org/10.26717/BJSTR.2019.16.002837>.
- [41] M.A. Fikiet, S.R. Khandasammy, E. Mistek, Y. Ahmed, L. Halámková, J. Bueno, et al., Surface enhanced Raman spectroscopy: a review of recent applications in forensic science, Spectrochim. Acta A Mol. Biomol. Spectrosc. 197 (2018) 255–260, <https://doi.org/10.1016/J.SAA.2018.02.046>.
- [42] B.K. Lavine, Chemometrics in forensic science, J. Chemom. 35 (2021) e3322, <https://doi.org/10.1002/CEM.3322>.
- [43] Chemometrics, Statistics and computer application in analytical chemistry, 37-0336-37-0336, Choice Rev. Online 37 (1999), <https://doi.org/10.5860/CHOICE.37-0336>.
- [44] Lewis I.R., Edwards H.G.M. Handbook of Raman spectroscopy: from the research laboratory to the process line 2001:1054.
- [45] C. Author, Forensic applications of Raman spectroscopy a review, J. Bioanal. Methods Tech. 2 (2022) 102.
- [46] S.R. Khandasammy, M.A. Fikiet, E. Mistek, Y. Ahmed, L. Halámková, J. Bueno, et al., Bloodstains, paintings, and drugs: Raman spectroscopy applications in forensic science, Forensic Chem. 8 (2018) 111–133, <https://doi.org/10.1016/J.FORC.2018.02.002>.
- [47] M. Claybourn, M. Ansell, Using Raman spectroscopy to solve crime: inks, questioned documents and fraud, Sci. Justice 40 (2000) 261–271, [https://doi.org/10.1016/S1355-0306\(00\)71996-4](https://doi.org/10.1016/S1355-0306(00)71996-4).
- [48] S. Chauhan, S. Sharma, Applications of Raman spectroscopy in the analysis of biological evidence, Forensic Sci. Med. Pathol. (2023), <https://doi.org/10.1007/S12024-023-00660-Z>.
- [49] D. Cialla-May, M. Schmitt, J. Popp, Theoretical principles of Raman spectroscopy, Phys. Sci. Rev. 4 (2019), <https://doi.org/10.1515/PSR-2017-0040>.
- [50] F. Madzharova, Z. Heiner, J. Kneipp, Surface enhanced hyper Raman scattering (SEHRS) and its applications, Chem. Soc. Rev. 46 (2017) 3980–3999, <https://doi.org/10.1039/C7CS00137A>.
- [51] G. McNay, D. Eustace, W.E. Smith, K. Faulds, D. Graham, Surface-enhanced Raman scattering (SERS) and surface-enhanced resonance Raman scattering (SERRS): a review of applications, Appl. Spectrosc. 65 (2011) 825–837, <https://doi.org/10.1366/11-06365>.
- [52] El-Mashtoly S.F. Confocal Raman Microscopy, in: Jan Toporski, Thomas Dieing, Olaf Hollricher (Eds.), Microscopy and Microanalysis 2019;25:1293–4, Second Edition, Springer, New York, 2018, p. 596, <https://doi.org/10.1017/S13431927619014880>. ISBN: 978-319753782.
- [53] C.H. Camp, Y.J. Lee, M.T. Cicerone, Quantitative, comparable coherent Anti-Stokes Raman scattering (CARS) spectroscopy: correcting errors in phase retrieval, J. Raman Spectrosc. 47 (2016) 408–415, <https://doi.org/10.1002/JRS.4824>.
- [54] J.J. Barrett, D.R. Siebert, G.A. West, Applications of photoacoustic Raman spectroscopy, NonLinear Raman Spectrosc. Chem. Apl. (1982) 585–602, https://doi.org/10.1007/978-94-009-7912-3_29.
- [55] R.W. Havener, H. Zhuang, L. Brown, R.G. Hennig, J. Park, Angle-resolved Raman imaging of interlayer rotations and interactions in twisted bilayer graphene, Nano Lett. 12 (2012) 3162–3167, https://doi.org/10.1021/NL301137K/SUPPL_FILE/NL301137K_SI_001.PDF.
- [56] I. Noda, A.E. Dowrey, C. Marcott, G.M. Story, Y. Ozaki, Generalized Two-Dimensional correlation spectroscopy, Appl. Spectrosc. 54 (2000), <https://doi.org/10.1366/0003702001950454>.
- [57] M. Sims, S.J. Jaret, J.R. Johnson, M.L. Whitaker, T.D. Glotch, Unconventional high-pressure Raman spectroscopy study of kinetic and peak pressure effects in plagioclase feldspars, Phys. Chem. Min. 47 (2020), <https://doi.org/10.1007/S00269-020-01080-Z>.
- [58] A.F. Goncharov, Raman spectroscopy at high pressures, Int. J. Spectrosc. 2012 (2012) 1–16, <https://doi.org/10.1155/2012/617528>.
- [59] X. Zhao, N. Wang, M. Zhu, X. Qiu, S. Sun, Y. Liu, et al., Application of transmission Raman spectroscopy in combination with partial Least-Squares (PLS) for the fast quantification of paracetamol, Mol. 2022 Vol. 27 Page 1707 27 (2022) 1707, <https://doi.org/10.3390/MOLECULES27051707>.
- [60] E. Villa-Aleman, A.L. Houk, D.D. Dick, S.E. Hunyadi Murph, Hyper-Raman spectroscopy of CeO₂, J. Raman Spectrosc. 51 (2020) 1260–1263, <https://doi.org/10.1002/JRS.5886>.
- [61] B. Robert, Resonance Raman spectroscopy, Photosynth Res 101 (2009) 147–155, <https://doi.org/10.1007/S1120-009-9440-4>.

- [62] S. Mosca, C. Conti, N. Stone, P. Matousek, Spatially offset Raman spectroscopy, *Nat. Rev. Methods Prim.* 2021 11 1 (2021) 1–16, <https://doi.org/10.1038/s43586-021-00019-0>.
- [63] S.K. Sahoo, S. Umaphathy, A.W. Parker, Time-resolved resonance Raman spectroscopy: exploring reactive intermediates, *Appl. Spectrosc.* 65 (2011) 1087–1115, <https://doi.org/10.1366/11-06406>.
- [64] P. Kukura, D.W. McCamant, R.A. Mathies, Femtosecond stimulated Raman spectroscopy, *Annu. Rev. Phys. Chem.* 58 (2007) 461–488, <https://doi.org/10.1146/ANNUREV.PHYSICHEM.58.032806.104456>.
- [65] A.J. Barnes, Infrared and Raman matrix isolation spectroscopy, *Matrix Isol. Spectrosc.* (1981) 13–26, https://doi.org/10.1007/978-94-009-8540-7_2.
- [66] L.D. Barron, The development of biomolecular Raman optical activity spectroscopy, *Biomed. Spectrosc. Imaging* 4 (2015) 223–253, <https://doi.org/10.3233/BSI-150113>.
- [67] S.O. Pour, L.D. Barron, S.T. Mutter, E.W. Blanch, Raman optical activity. Chiral analysis: advances in spectroscopy, *Chromatogr. Emerg. Methods*. Second Ed. (2018) 249–291, <https://doi.org/10.1016/B978-0-444-64027-7.00006-9>.
- [68] H.F. Ma, Y. Zhang, A.P. Ye, Single-cell discrimination based on optical tweezers Raman spectroscopy, *Chin. Sci. Bull.* 58 (2013) 2594–2600, <https://doi.org/10.1007/S11434-013-5721-6>.
- [69] C.E. Buffett, M.D. Morris, Rapid scan inverse Raman spectroscopy, *Appl. Spectrosc.* 35 (1981) 203–205, <https://doi.org/10.1366/0003702814731699>.
- [70] T.T. Basiev, A.A. Sobol, Y.K. Voronko, P.G. Zverev, Spontaneous Raman spectroscopy of tungstate and molybdate crystals for Raman lasers, *Opt. Mater.* (Amst.) 15 (2000) 205–216, [https://doi.org/10.1016/S0925-3467\(00\)00037-9](https://doi.org/10.1016/S0925-3467(00)00037-9).
- [71] P. Verma, Tip-Enhanced Raman spectroscopy: technique and recent advances, *Chem. Rev.* 117 (2017) 6447–6466, <https://doi.org/10.1021/ACS.CHEMREV.6B00821/ASSET/IMAGES/MEDIUM/CR-2016-00821H.0011.GIF>.
- [72] P. Buzzini, G. Massonnet, F.M. Sermier, The micro Raman analysis of paint evidence in criminalistics: case studies, *J. Raman Spectrosc.* 37 (2006) 922–931, <https://doi.org/10.1002/JRS.1522>.
- [73] G. Jochem, R.J. Lehnert, On the potential of Raman microscopy for the forensic analysis of coloured textile fibres, *Sci. Justice* 42 (2002) 215–221, [https://doi.org/10.1016/S1355-0306\(02\)71831-5](https://doi.org/10.1016/S1355-0306(02)71831-5).
- [74] F. Zapata, C. García-Ruiz, Analysis of different materials subjected to open-air explosions in search of explosive traces by Raman microscopy, *Forensic Sci. Int.* 275 (2017) 57–64, <https://doi.org/10.1016/J.FORSINT.2017.02.032>.
- [75] F. Zapata, C. García-Ruiz, Determination of nanogram microparticles from explosives after real Open-Air explosions by confocal Raman microscopy, *Anal. Chem.* 88 (2016) 6726–6733, <https://doi.org/10.1021/ACS.ANALCHEM.6B00927>.
- [76] S. Gomes da Costa, A. Richter, U. Schmidt, S. Breuninger, O. Hollricher, Confocal Raman microscopy in life sciences, *Morphologie* 103 (2019) 11–16, <https://doi.org/10.1016/J.MORPHO.2018.12.003>.
- [77] Toporski J., Dieing T., Hollricher O., editors. *Confocal Raman Microscopy* 2018; 66. <https://doi.org/10.1007/978-3-319-75380-5>.
- [78] M. Ogawa, Y. Harada, Y. Yamaoka, K. Fujita, H. Yaku, T. Takamatsu, Label-free biochemical imaging of heart tissue with high-speed spontaneous Raman microscopy, *Biochem Biophys. Res. Commun.* 382 (2009) 370–374, <https://doi.org/10.1016/J.BBRC.2009.03.028>.
- [79] A.F. Palonpon, M. Sodeoka, K. Fujita, Molecular imaging of live cells by Raman microscopy, *Curr. Opin. Chem. Biol.* 17 (2013) 708–715, <https://doi.org/10.1016/J.CBPA.2013.05.021>.
- [80] M.K. Purba, S. Gupta, R.R. Gupta, S.K. Shukla, A survey of techniques used for ink examination, *J. Punjab. Acad. Forensic Med. Toxicol.* 19 (2019) 155–161, <https://doi.org/10.5958/0974-083X.2019.00032.3>.
- [81] M.S. Božičević, A. Gajović, I. Zjakić, Identifying a common origin of toner printed counterfeit banknotes by micro-Raman spectroscopy, *Forensic Sci. Int.* 223 (2012) 314–320, <https://doi.org/10.1016/J.FORSINT.2012.10.007>.
- [82] R. Saferstein, A.B. Hall. *Forensic science handbook: Third edition, Third Edition, Forensic Science Handbook*, 2017, pp. 1–761, <https://doi.org/10.4324/9781315119939>.
- [83] (PDF) Application of Raman Spectroscopy and Surface-Enhanced Raman Scattering to the Analysis of Synthetic Dyes Found in Ballpoint Pen Inks n.d. (https://www.researchgate.net/publication/24441386_Application_of_Raman_Spectroscopy_and_Surface-Enhanced_Raman_Scattering_to_the_Analysis_of_Synthetic_Dyes_Found_in_Ballpoint_Pen_Inks) (accessed July 11, 2022).
- [84] E.G. Udriștioiu, A.A. Bunaciu, H.Y. Aboul-Enein, I.G. Tănase, Forensic analysis of color toners by Raman spectroscopy, *Instrum. Sci. Technol.* 37 (2009) 23–29, <https://doi.org/10.1080/10739140802584707>.
- [85] M.R. de Almeida, D.N. Correa, W.F.C. Rocha, F.J.O. Scaff, R.J. Poppi, Discrimination between authentic and counterfeit banknotes using Raman spectroscopy and PLS-DA with uncertainty estimation, *Microchem. J.* 109 (2013) 170–177, <https://doi.org/10.1016/J.MICROC.2012.03.006>.
- [86] L. Heudt, D. Debois, T.A. Zimmerman, L. Köhler, F. Bano, F. Partouche, et al., Raman spectroscopy and laser desorption mass spectrometry for minimal destructive forensic analysis of black and color inkjet printed documents, *Forensic Sci. Int.* 219 (2012) 64–75, <https://doi.org/10.1016/J.FORSINT.2011.12.001>.
- [87] W.W. Yu, L.M. White, Inkjet printed surface enhanced Raman spectroscopy array on cellulose paper, *Anal. Chem.* 82 (2010) 9626–9630, https://doi.org/10.1021/AC102475K/ASSET/IMAGES/LARGE/AC-2010-02475K_0004.JPEG.
- [88] S.E.J. Bell, S.P. Stewart, Y.C. Ho, B.W. Craythorne, S.J. Speers, Comparison of the discriminating power of Raman and surface-enhanced Raman spectroscopy with established techniques for the examination of liquid and gel inks, *J. Raman Spectrosc.* 44 (2013) 509–517, <https://doi.org/10.1002/JRS.4202>.
- [89] R. Chen, J. Lv, J. Feng, Y. Liu, W. Zhang, Discrimination of seal inks used for seals by confocal Raman microscopy, *Pigment Resin Technol.* 43 (2014) 389–393, <https://doi.org/10.1108/PRT-10-2013-0096>.
- [90] S. Liu, J. Feng, J. Lv, W. Zhang, Discrimination of blue ballpoint pen inks in Chinese market with confocal Raman microscope, *Pigment Resin Technol.* 43 (2014) 45–51, <https://doi.org/10.1108/PRT-12-2012-0089/FULL/XML>.
- [91] L.C. Lee, M.I.A. Samad, M.A.M. Ismail, Nondestructive classification and identification of ballpoint pen inks by Raman spectroscopy for forensic document examinations, *J. Anal. Chem.* 2016 717 71 (2016) 723–729, <https://doi.org/10.1134/S106193481607011X>.
- [92] M.I. Nunkoo, M.B. Saib-Sunassy, H. Li Kam Wah, S. Jhaumeer Laulloo, Forensic analysis of black, blue, red, and Green ballpoint pen inks, *Cryst. Ideas Role Chem.* (2016) 323–339, https://doi.org/10.1007/978-3-319-31759-5_21.
- [93] K.O. Gorshkova, I.I. Tumkin, L.A. Myund, A.S. Tverjanovich, A.S. Mereshchenko, M.S. Panov, et al., The investigation of dye aging dynamics in writing inks using Raman spectroscopy, *Dyes Pigments* 131 (2016) 239–245, <https://doi.org/10.1016/J.DYEPIG.2016.04.009>.
- [94] F.D.S.L. Borba, T. Jawhari, R. Saldanha Honorato, A. De Juan, Confocal Raman imaging and chemometrics applied to solve forensic document examination involving crossed lines and obliteration cases by a depth profiling study, *Analyst* 142 (2017) 1106–1118, <https://doi.org/10.1039/C6AN02340A>.
- [95] S.I. Sharaa, A.A.A. El-Magd, A.A. Bakr, Y.M. Moustafa, I.M.A. El-Aziz, Utilizing the combination of a digital microscope and Raman spectroscopy in determination the sequence of intersecting strokes from laser toner and water-based ink, *Dyes Pigments* 163 (2019) 393–411, <https://doi.org/10.1016/J.DYEPIG.2018.12.016>.
- [96] L. Rodrigues e Brito, A.B. Chaves, A. Braz, M.F. Pimentel, Raman hyperspectral imaging and a novel approach for objective determination of the order of crossing ink lines, *Spectrochim. Acta A Mol. Biomol. Spectrosc.* 223 (2019), <https://doi.org/10.1016/J.SAA.2019.117287>.
- [97] M.N. Mohamad Asri, R. Verma, N.A. Mahat, N.A.M. Nor, W.N.S. Mat Desa, D. Ismail, Discrimination and source correspondence of black gel inks using Raman spectroscopy and chemometric analysis with UMAP and PLS-DA, *Chemom. Intell. Lab. Syst.* 225 (2022) 104557, <https://doi.org/10.1016/J.CHEMOLAB.2022.104557>.
- [98] S.T. Saravanan, J. Ganesamurthi, S.-M. Chen, T.-W. Chen, C.-J. Chen, K. Lakshmanan, et al., Non-Destructive discrimination of blue inks on suspected documents through the combination of Raman spectroscopy and chemometric analysis, *J. (Basel)* 6 (2023) 536–543, <https://doi.org/10.3390/J6040035>.
- [99] C.G. Sarkar, Pigments in ancient manuscripts and paintings brought to life by Raman spectroscopy: a short review, *Orient. J. Chem.* 38 (2022) 1081–1093, <https://doi.org/10.13005/OJC/380501>.
- [100] A. Ferretti, I. Degano, S. Legnaioli, B. Campanella, A. Sainati, M.P. Colombini, Shedding light on the composition and degradation mechanism of dyes in historical ink's collection (19th–20th century), *Dyes Pigments* 220 (2023) 111672, <https://doi.org/10.1016/J.DYEPIG.2023.111672>.
- [101] D. Su, K. Li, Micro Raman spectroscopy combined with chromatographic effect and OPLS-DA for time-resolved detection of stamp inks, *Microchem. J.* 190 (2023) 108727, <https://doi.org/10.1016/J.MICROC.2023.108727>.
- [102] J. Zieba-Palus, R. Borusiewicz, M. Kunicki, PRAXIS-combined mu-Raman and mu-XRF spectrometers in the examination of forensic samples, *Forensic Sci. Int.* 175 (2008) 1–10, <https://doi.org/10.1016/J.FORSINT.2007.04.230>.
- [103] K.W.C. Poon, I.R. Dadour, A.J. McKinley, In situ chemical analysis of modern organic tattooing inks and pigments by micro-Raman spectroscopy, *J. Raman Spectrosc.* 39 (2008) 1227–1237, <https://doi.org/10.1002/JRS.1973>.
- [104] A.S. Lee, V. Otieno-Alego, D.C. Creagh, Identification of iron-gall inks with near-infrared Raman microspectroscopy, *J. Raman Spectrosc.* 39 (2008) 1079–1084, <https://doi.org/10.1002/JRS.1989>.
- [105] I. Geiman, M. Leona, J.R. Lombardi, Application of Raman spectroscopy and surface-enhanced Raman scattering to the analysis of synthetic dyes found in ballpoint pen inks, *J. Forensic Sci.* 54 (2009) 947–952, <https://doi.org/10.1111/J.1556-4029.2009.01058.X>.
- [106] Development of Advanced Raman Spectroscopy Methods and Databases for the Evaluation of Trace Evidence and the Examination of Questioned Documents-Phase II | National Institute of Justice n.d. (<https://nij.ojp.gov/library/publications/development-advanced-raman-spectroscopy-methods-and-databases-evaluation-0>) (accessed November 30, 2022).
- [107] L.C. Abbott, S.N. Batchelor, J.R.L. Smith, J.N. Moore, Resonance Raman and UV-visible spectroscopy of black dyes on textiles, *Forensic Sci. Int.* 202 (2010) 54–63, <https://doi.org/10.1016/J.FORSINT.2010.04.026>.
- [108] Colombini A., Kaifas D. CHARACTERIZATION OF SOME ORANGE AND YELLOW ORGANIC AND FLUORESCENT PIGMENTS BY RAMAN SPECTROSCOPY 2010.
- [109] M. Hoehse, A. Paul, I. Gornushkin, U. Panne, Multivariate classification of pigments and inks using combined Raman spectroscopy and LIBS, *Anal. Bioanal. Chem.* 402 (2012) 1443–1450, <https://doi.org/10.1007/S00216-011-5287-6>.
- [110] X.F. Wang, J. Yu, A.L. Zhang, D.W. Zhou, M.X. Xie, Nondestructive identification for red ink entries of seals by Raman and Fourier transform infrared spectrometry, *Spectrochim. Acta A Mol. Biomol. Spectrosc.* 97 (2012) 986–994, <https://doi.org/10.1016/J.SAA.2012.07.127>.
- [111] A. Raza, B. Saha, Application of Raman spectroscopy in forensic investigation of questioned documents involving stamp inks, *Sci. Justice* 53 (2013) 332–338, <https://doi.org/10.1016/J.SCIJUS.2012.11.001>.
- [112] A. Raza, B. Saha, Silver nanoparticles doped agarose disk: highly sensitive surface-enhanced Raman scattering substrate for in situ analysis of ink dyes, *Forensic Sci. Int.* 233 (2013) 21–27, <https://doi.org/10.1016/J.FORSINT.2013.08.004>.

- [113] (PDF) A Comparison of Two Raman Spectrometer Systems for the Analysis of Gel Pen Ink Writings on White Office Paper Using 785 nm Excitation Wavelength n.d. (https://www.researchgate.net/publication/260083347_A_Comparison_of_Two_Raman_Spectrometer_Systems_for_the_Analysis_of_Gel_Pen_Ink_Writings_on_White_Office_Paper_Using_785_nm_Excitation_Wavelength) (accessed November 30, 2022).
- [114] M. Król, A. Karoly, P. Kościelniak, Raman spectroscopy and capillary electrophoresis applied to forensic colour inkjet printer inks analysis, *Forensic Sci. Int.* 242 (2014) 142–149, <https://doi.org/10.1016/J.FORSINT.2014.06.031>.
- [115] A. Braz, M. López-López, C. García-Ruiz, Studying the variability in the Raman signature of writing pen inks, *Forensic Sci. Int.* 245 (2014) 38–44, <https://doi.org/10.1016/J.FORSINT.2014.10.014>.
- [116] D.P. Benedetti, J. Zhang, T.J. Tague, J.R. Lombardi, M. Leona, In situ microanalysis of organic colorants by inkjet colloid deposition surface-enhanced Raman scattering, *J. Raman Spectrosc.* 45 (2014) 123–127, <https://doi.org/10.1002/JRS.4424>.
- [117] Braz A., López-López M., Montalvo G., Ruiz C.G. Forensic discrimination of inkjet-printed lines by Raman spectroscopy and surface-enhanced Raman spectroscopy. <http://DxDoiOrg/101080/004506182014982179> 2014;47:411–420. <https://doi.org/10.1080/00450618.2014.982179>.
- [118] F. de Souza Lins Borba, R. Saldanha Honorato, A. de Juan, Use of Raman spectroscopy and chemometrics to distinguish blue ballpoint pen inks, *Forensic Sci. Int.* 249 (2015) 73–82, <https://doi.org/10.1016/J.FORSINT.2015.01.027>.
- [119] A. Braz, M. López-López, C. García-Ruiz, Raman imaging for determining the sequence of blue pen ink crossings, *Forensic Sci. Int.* 249 (2015) 92–100, <https://doi.org/10.1016/J.FORSINT.2015.01.023>.
- [120] V. Vančo, M. Kadlecíková, J. Breza, P. Michniak, M. Štěpánek, M. Reháková, et al., Differentiation of selected blue writing inks by surface-enhanced Raman spectroscopy, *Chem. Pap.* 69 (2015) 518–526, <https://doi.org/10.1515/CHEMPAP-2015-0068>.
- [121] D. Kuroski, R.P. Van Duyn, In situ detection and identification of hair dyes using surface-enhanced Raman spectroscopy (SERS), *Anal. Chem.* 87 (2015) 2901–2906, <https://doi.org/10.1021/AC504405U>.
- [122] P. Buzzini, G. Massonnet, The analysis of colored acrylic, cotton, and wool textile fibers using micro-Raman spectroscopy. Part 2: comparison with the traditional methods of fiber examination, *J. Forensic Sci.* 60 (2015) 712–720, <https://doi.org/10.1111/1556-4029.12654>.
- [123] E.A. Carter, F.R. Perez, J.M. Garcia, H.G.M. Edwards, Raman spectroscopic analysis of an important visigothic historiated manuscript, *Philos. Trans. A Math. Phys. Eng. Sci.* 374 (2016), <https://doi.org/10.1098/RSTA.2016.0041>.
- [124] M.N. Mohamad Asri, W.N.S. Mat Desa, D. Ismail, Raman spectroscopy of ballpoint-pen inks using chemometric techniques, *Aust. J. Forensic Sci.* 49 (2017) 175–185, <https://doi.org/10.1080/00450618.2016.1153712>.
- [125] N.M. Grechukha, K.O. Gorshkova, M.S. Panov, I.I. Tumkin, E.O. Kirillova, V. V. Lukianov, et al., Analysis of the aging processes of writing ink: Raman spectroscopy versus gas chromatography aspects, *Appl. Sci.* 2017 Vol. 7 Page 991 7 (2017) 991, <https://doi.org/10.3390/AP7100991>.
- [126] K.O. Gorshkova, I.I. Tumkin, E.O. Kirillova, M.S. Panov, V.A. Kochemirovsky, K.O. Gorshkova, et al., The development of methods of analysis of documents on the basis of the methods of Raman spectroscopy and fluorescence analysis, *SPIE* 10173 (2017) 1017313, <https://doi.org/10.1117/12.2268349>.
- [127] A. Cesaratto, J.R. Lombardi, M. Leona, Tracking photo-degradation of triarylmethane dyes with surface-enhanced Raman spectroscopy, *J. Raman Spectrosc.* 48 (2017) 418–424, <https://doi.org/10.1002/JRS.5056>.
- [128] C.E. Johnson, P. Martin, K.A. Roberts, T. Trejos, R. Corzo, J.R. Almirall, et al., The capability of Raman microspectroscopy to differentiate printing inks, *J. Forensic Sci.* 63 (2018) 66–79, <https://doi.org/10.1111/1556-4029.13508>.
- [129] M.N. Mohamad Asri, W.N.S. Mat Desa, D. Ismail, Source determination of red gel pen inks using Raman spectroscopy and attenuated total reflectance Fourier transform infrared spectroscopy combined with Pearson's product moment correlation coefficients and principal component analysis, *J. Forensic Sci.* 63 (2018) 285–291, <https://doi.org/10.1111/1556-4029.13522>.
- [130] P. Buzzini, C. Polston, M. Schackmuth, On the criteria for the discrimination of inkjet printer inks using micro-Raman spectroscopy, *J. Raman Spectrosc.* 49 (2018) 1791–1801, <https://doi.org/10.1002/JRS.5458>.
- [131] D. Saviello, M. Trabace, A. Alyami, A. Mirabile, R. Giorgi, P. Baglioni, et al., A combined surface enhanced Raman spectroscopy (SERS)/UV–vis approach for the investigation of dye content in commercial felt tip pens inks, *Talanta* 181 (2018) 448–453, <https://doi.org/10.1016/J.TALANTA.2018.01.049>.
- [132] A.R. Novais Rodrigues, F.L. Melquiades, C.R. Appoloni, E.N. Marques, Characterization of Brazilian banknotes using portable X-ray fluorescence and Raman spectroscopy, *Forensic Sci. Int.* 302 (2019), <https://doi.org/10.1016/J.FORSINT.2019.06.030>.
- [133] M.N.M. Asri, N.N. Yahya, N.A.M. Nor, W.N.S.M. Desa, D. Ismail, Towards establishing a non-destructive technique for forensic ink analysis involving Raman spectroscopy with chemometric procedures, *AIP Conf. Proc.* 2155 (2019) 020006, <https://doi.org/10.1063/1.5125510>.
- [134] D. Saviello, M. Trabace, A. Alyami, A. Mirabile, P. Baglioni, R. Giorgi, et al., Raman spectroscopy and surface enhanced Raman scattering (SERS) for the analysis of blue and black writing inks: identification of dye content and degradation processes, *Front Chem.* 7 (2019), <https://doi.org/10.3389/FCHEM.2019.00727>.
- [135] C.A. Teixeira, R.J. Poppi, Discriminating blue ballpoint pens inks in questioned documents by Raman imaging and mean-field approach independent component analysis (MFICA), *Microchem. J.* 144 (2019) 411–418, <https://doi.org/10.1016/J.MICROC.2018.10.002>.
- [136] K.O. Gorshkova, E.R. Rossinskaya, N.P. Kirillova, A.A. Fogel, S. V. Kochemirovskaya, V.A. Kochemirovsky, Investigation of the new possibility of mathematical processing of Raman spectra for dating documents, *Sci. Justice* 60 (2020) 451–465, <https://doi.org/10.1016/J.SCIJUS.2020.06.007>.
- [137] N. Nuntawong, S. Limwichean, M. Horprathum, V. Patthanasettakul, A. Ketkong, K. Thamaphat, et al., Improved discrimination of pen inks on document by surface-enhanced Raman substrate fabricated by magnetron sputtering, *Opt. (Stuttg.)* 201 (2020) 163499, <https://doi.org/10.1016/J.JLEO.2019.163499>.
- [138] V. Ramanathan, L. Keseven, Raman spectroscopy in forensic analysis of blue gel inks on paper, *Proceedings Indian National Science Academy (Online)* 85 (2019) 613–618, <https://doi.org/10.16943/PTINSA/2019/49581>.
- [139] M.N.M. Asri, N.F. Nestrgan, N.A.M. Nor, R. Verma, On the discrimination of inkjet, laser and photocopier printed documents using Raman spectroscopy and chemometrics: application in forensic science, *Microchem. J.* 165 (2021) 106136, <https://doi.org/10.1016/J.MICROC.2021.106136>.
- [140] P. Buzzini, J. Curran, C. Polston, Comparison between visual assessments and different variants of linear discriminant analysis to the classification of Raman patterns of inkjet printer inks, *Forensic Chem.* 24 (2021) 100336, <https://doi.org/10.1016/J.FORC.2021.100336>.
- [141] M. Naeim Mohamad Asri, R. Verma, N. Arafat Mahat, N. Azman Mohd Nor, W. Nur Syuhaila Mat Desa, D. Ismail, Raman spectroscopy with self-organizing feature maps and partial least squares discriminant analysis for discrimination and source correspondence of red gel ink pens, *Microchem. J.* 175 (2022) 107170, <https://doi.org/10.1016/J.MICROC.2021.107170>.
- [142] C. Sessa, C. Steuer, D. Quintero Balbas, G. Scutti, S. Prati, H. Stege, Analytical studies on commercial artists' colour charts from das deutsche farbenbuch (1925)—identification of synthetic and natural organic colourants by Raman microscopy, surface-enhanced Raman spectroscopy and metal underlayer ATR-FTIR spectroscopy, *Herit. Sci.* 10 (2022), <https://doi.org/10.1186/S40494-022-00740-3>.
- [143] S. Higgins, D. Kuroski, Surface-enhanced Raman spectroscopy enables highly accurate identification of different brands, types and colors of hair dyes, *Talanta* 251 (2023) 123762, <https://doi.org/10.1016/J.TALANTA.2022.123762>.
- [144] M.M. Eisnor, K.E.R. McLeod, S. Bindsri, S. Svoboda, K.L. Wustholz, C. L. Brosseau, Electrochemical surface-enhanced Raman spectroscopy (EC-SERS): a tool for the identification of polyphenolic components in natural lake pigments, *Phys. Chem. Phys.* 24 (2021) 347–356, <https://doi.org/10.1039/D1CP03301H>.
- [145] A. Ciccola, K.R. McClure, I. Serafini, F. Vincenti, C. Montesano, A. Gentili, et al., The 20th century and its new colours: investigating the molecular structures of historical synthetic dyes using Raman spectroscopy, *J. Raman Spectrosc.* 55 (2024) 324–335, <https://doi.org/10.1002/JRS.6624>.
- [146] A. Braz, M. López-López, C. García-Ruiz, Raman spectroscopy for forensic analysis of inks in questioned documents, *Forensic Sci. Int.* 232 (2013) 206–212, <https://doi.org/10.1016/J.FORSINT.2013.07.017>.
- [147] K. Retko, L. Legan, J. Kosel, P. Ropret, Identification of iron gall inks, logwood inks, and their mixtures using Raman spectroscopy, supplemented by reflection and transmission infrared spectroscopy, *Herit. Sci.* 12 (2024), <https://doi.org/10.1186/S40494-024-01323-0>.
- [148] C. Guo, M. Hao, S. Liu, J. Lv, Forensic evaluation of ink age by Raman spectroscopy and gas Chromatography-Mass spectrometry (GC-MS), *Anal. Lett.* (2024), <https://doi.org/10.1080/00032719.2024.2333959>.
- [149] Y. Zou, A. Zhang, X. Wang, L. Yang, M. Ding, Comparison of feature selection and data fusion of Fourier transform infrared and Raman spectroscopy for identifying watercolor ink, *J. Forensic Sci.* 69 (2024) 584–592, <https://doi.org/10.1111/1556-4029.15468>.
- [150] Karger B. *Forensic Ballistics* 2009:139–172. https://doi.org/10.1007/978-1-59745-110-9_9.
- [151] O. Dalby, D. Butler, J.W. Birkett, Analysis of gunshot residue and associated materials—a review, *J. Forensic Sci.* 55 (2010) 924–943, <https://doi.org/10.1111/J.1556-4029.2010.01370.X>.
- [152] V.R. Minzière, A.-L. Gassner, M. Gallidabino, C. Roux, Céline Weyermann, The relevance of gunshot residues in forensic science, *Wiley Interdisciplinary Reviews Forensic Science* 5 (2023) e1472, <https://doi.org/10.1002/WFS2.1472>.
- [153] P. Shrivastava, V.K. Jain, S. Nagpal, Gunshot residue detection technologies—a review, *Egypt. J. Forensic Sci.* 2021 111 11 (2021) 1–21, <https://doi.org/10.1186/S41935-021-00223-9>.
- [154] J. Bueno, V. Sikirzhyski, I.K. Lednev, Raman spectroscopic analysis of gunshot residue offering great potential for caliber differentiation, *Anal. Chem.* 84 (2012) 4334–4339, <https://doi.org/10.1021/AC203429X>.
- [155] J. Bueno, I.K. Lednev, Advanced statistical analysis and discrimination of gunshot residue implementing combined Raman and FT-IR data, *Anal. Methods* 5 (2013) 6292–6296, <https://doi.org/10.1039/C3AY40721G>.
- [156] M. López-López, J.J. Delgado, C. García-Ruiz, Ammunition identification by means of the organic analysis of gunshot residues using Raman spectroscopy, *Anal. Chem.* 84 (2012) 3581–3585, <https://doi.org/10.1021/AC203237W>.
- [157] M. López-López, V. Merk, C. García-Ruiz, J. Kneipp, Surface-enhanced Raman spectroscopy for the analysis of smokeless gunpowders and macroscopic gunshot residues, *Anal. Bioanal. Chem.* 2016 40818 408 (2016) 4965–4973, <https://doi.org/10.1007/S00216-016-9591-Z>.
- [158] C.R. Carneiro, C.S. Silva, M.A. De Carvalho, M.F. Pimentel, M. Talhavi, I. T. Weber, Identification of luminescent markers for gunshot residues: fluorescence, Raman spectroscopy, and chemometrics, *Anal. Chem.* 91 (2019) 12444–12452, https://doi.org/10.1021/ACS.ANALCHEM.9B03079/SUPPL_FILE/AC9B03079_SI_001.PDF.

- [159] E. Thayer, W. Turner, S. Blama, M.S. Devadas, E.M. Hondrogiannis, Signal detection limit of a portable Raman spectrometer for the SERS detection of gunshot residue, *MRS Commun.* 9 (2019) 948–955, <https://doi.org/10.1557/MRC.2019.100>.
- [160] B. Jain, P. Yadav, Vibrational spectroscopy and chemometrics in GSR: review and current trend, *Egypt J. Forensic Sci.* 11 (2021), <https://doi.org/10.1186/S41935-021-00229-3>.
- [161] S.R. Khandasammy, N.R. Bartlett, L. Halámková, I.K. Lednev, Hierarchical modelling of Raman spectroscopic data demonstrates the potential for manufacturer and caliber differentiation of smokeless powders, *Chemosensors* 11 (2023), <https://doi.org/10.3390/CHEMOSENSORS11010011/S1>.
- [162] S.R. Khandasammy, L. Halámková, M. Baudelet, I.K. Lednev, Identification and highly selective differentiation of organic gunshot residues utilizing their elemental and molecular signatures, *Spectrochim. Acta A Mol. Biomol. Spectrosc.* 291 (2023), <https://doi.org/10.1016/J.SAA.2023.122316>.
- [163] American Academy of Forensic Sciences n.d. (<https://www.aafs.org/research/detection-organic-components-gunshot-residue-gsr-carbon-sem-stubs-raman-spectroscopy>) (accessed November 30, 2022).
- [164] Z. Abrego, N. Grijalba, N. Unceta, M. Maguregui, A. Sanchez, A. Fernández-Isla, et al., A novel method for the identification of inorganic and organic gunshot residue particles of lead-free ammunitions from the hands of shooters using scanning laser ablation-ICPMS and Raman micro-spectroscopy, *Analyst* 139 (2014) 6232–6241, <https://doi.org/10.1039/C4AN01051E>.
- [165] J. Zeng, J. Qi, F. Bai, J.C. Chung Yu, W.C. Shih, Analysis of ethyl and methyl centralite vibrational spectra for mapping organic gunshot residues, *Analyst* 139 (2014) 4270–4278, <https://doi.org/10.1039/C4AN00657G>.
- [166] A.Y. Panarin, I.A. Khodasevich, O.L. Gladkova, S.N. Terekhov, Determination of antimony by surface-enhanced Raman spectroscopy, *Appl. Spectrosc.* 68 (2014) 297–306, <https://doi.org/10.1366/13-07034>.
- [167] M. Lopez-Lopez, M.A.F. De La Ossa, C. Garcia-Ruiz, Fast analysis of complete macroscopic gunshot residues on substrates using Raman imaging, *Appl. Spectrosc.* 69 (2015) 889–893, <https://doi.org/10.1366/14-07816>.
- [168] D.S. Karahacane, A. Dahmani, K. Khimeche, Raman spectroscopy analysis and chemometric study of organic gunshot residues originating from two types of ammunition, *Forensic Sci. Int.* 301 (2019) 129–136, <https://doi.org/10.1016/J.FORSCIINT.2019.05.022>.
- [169] S.R. Khandasammy, A. Rzhetskii, I.K. Lednev, A novel Two-Step method for the detection of organic gunshot residue for forensic purposes: fast fluorescence imaging followed by Raman microspectroscopy identification, *Anal. Chem.* 91 (2019) 11731–11737, https://doi.org/10.1021/ACS.ANALCHEM.9B02306/SUPPL_FILE/AC9B02306_SI_001.PDF.
- [170] T. Shafirovich, D. Aligholizadeh, M. Johnson, E. Hondrogiannis, M.S. Devadas, Point-and-shoot: portable Raman and SERS detection of organic gunshot residue analytes, *Vib. Spectrosc.* 131 (2024) 103669, <https://doi.org/10.1016/J.VIBSPEC.2024.103669>.
- [171] Forensic Examination of Glass and Paint. Forensic Examination of Glass and Paint 2001. <https://doi.org/10.1201/9780203483589/FORENSIC-EXAMINATION-GLASS-PAINT-BRIAN-CADDY>.
- [172] Basic components of automotive paint n.d.
- [173] Kobilinsky L.F. Forensic chemistry handbook 2012:504.
- [174] (PDF) Forensic examination of car paints n.d. https://www.researchgate.net/publication/294871293_Forensic_examination_of_car_paints (accessed July 14, 2022).
- [175] J. Ziba-Palus, J. Ws-Gubala, An investigation into the use of micro-Raman spectroscopy for the analysis of car paints and single textile fibres, *J. Mol. Struct.* 993 (2011) 127–133, <https://doi.org/10.1016/J.MOLSTRUC.2010.12.041>.
- [176] J. Ziba-Palus, A. Michalska, A. Weselucha-Birczyńska, Characterisation of paint samples by infrared and Raman spectroscopy for criminalistic purposes, *J. Mol. Struct.* 993 (2011) 134–141, <https://doi.org/10.1016/J.MOLSTRUC.2010.12.016>.
- [177] C. Muehlethaler, G. Massonnet, P. Esseiva, The application of chemometrics on infrared and Raman spectra as a tool for the forensic analysis of paints, *Forensic Sci. Int.* 209 (2011) 173–182, <https://doi.org/10.1016/J.FORSCIINT.2011.01.025>.
- [178] S.E.J. Bell, S.P. Stewart, W.J. Armstrong, Raman spectroscopy for forensic analysis of household and automotive paints, *Infrared Raman Spectrosc.* 51 (2012) 121–135, <https://doi.org/10.1002/9781119962328.CH4B>.
- [179] Raman Spectroscopy of Automotive and Architectural Paints: In situ Pigment Identification and Evidentiary Significance | Office of Justice Programs n.d. (<https://www.ojp.gov/ncjrs/virtual-library/abstracts/raman-spectroscopy-automotive-and-architectural-paints-situ-pigment>) (accessed July 22, 2022).
- [180] J. Zieba-Palus, A. Michalska, Characterization of blue pigments used in automotive paints by Raman spectroscopy, *J. Forensic Sci.* 59 (2014) 943–949, <https://doi.org/10.1111/1556-4029.12499>.
- [181] J.G. Lv, S. Liu, J.M. Feng, Y. Liu, S.D. Zhou, R. Chen, Effective identification of paints pigments in hit-and-run cases with confocal Raman microscope, *Pigment Resin Technol.* 45 (2016) 294–300, <https://doi.org/10.1108/PRT-05-2015-0044>.
- [182] Lv J., Zhang W., Liu S., Chen R., Feng J., Zhou S., et al. Analysis of 52 automotive coating samples for forensic purposes with Fourier transform infrared spectroscopy (FTIR) and Raman microscopy. <http://DxDoiOrg/101080/1527592220151091403> 2016;17:59–67. <https://doi.org/10.1080/15275922.2015.1091403>.
- [183] A.G. Gomes De Oliveira, E. Wiercigroch, J. De Andrade Gomes, K. Malek, Infrared and Raman spectroscopy of automotive paints for forensic identification of natural weathering, *Anal. Methods* 10 (2018) 1203–1212, <https://doi.org/10.1039/C7AY02684F>.
- [184] D. Lambert, C. Muehlethaler, P. Esseiva, G. Massonnet, Combining spectroscopic data in the forensic analysis of paint: application of a multiblock technique as chemometric tool, *Forensic Sci. Int.* 263 (2016) 39–47, <https://doi.org/10.1016/J.FORSCIINT.2016.03.049>.
- [185] J. Zieba-Palus, Examination of the variation of chemical composition and structure of paint within a car body by FT-IR and Raman spectroscopies, *J. Mol. Struct.* 1219 (2020) 128558, <https://doi.org/10.1016/J.MOLSTRUC.2020.128558>.
- [186] Buzzini P., Massonnet G., Sermier F.M., Hafsi S., Ranzi R., Gueissaz L., et al. The Interpretation of Paint Evidence through the Use of Population Studies: Two Case Examples 2007.
- [187] M. Skenderovska, B. Minčeva-Šukarova, L. Andreeva, Application of Micro-Raman and FT-IR spectroscopy in forensic analysis of automotive topcoats in the republic of Macedonia, *Maced. J. Chem. Chem. Eng.* 27 (2008) 9–17, <https://doi.org/10.20450/MJCCE.2008.240>.
- [188] A. Hernanz, I. Bratu, O.F. Marutoiu, C. Marutoiu, J.M. Gavira-Vallejo, H.G. M. Edwards, Micro-Raman spectroscopic investigation of external wall paintings from st. Dumitru's Church, suceava, Romania, *Anal. Bioanal. Chem.* 392 (2008) 263–268, <https://doi.org/10.1007/S00216-008-2262-Y>.
- [189] S.P. Stewart, S.E.J. Bell, W.J. Armstrong, G. Kee, S.J. Speers, Forensic examination of multilayer White paint by lateral scanning Raman spectroscopy, *J. Raman Spectrosc.* 43 (2012) 131–137, <https://doi.org/10.1002/JRS.2982>.
- [190] J. Zieba-Palus, B.M. Trzcińska, Application of infrared and Raman spectroscopy in paint trace examination, *J. Forensic Sci.* 58 (2013) 1359–1363, <https://doi.org/10.1111/1556-4029.12183>.
- [191] A. Martyna, A. Michalska, G. Zadora, Interpretation of FTIR spectra of polymers and Raman spectra of car paints by means of likelihood ratio approach supported by wavelet transform for reducing data dimensionality, *Anal. Bioanal. Chem.* 407 (2015) 3357–3376, <https://doi.org/10.1007/S00216-015-8558-9>.
- [192] G. Germinario, I.D. van der Werf, L. Sabbatini, Chemical characterisation of spray paints by a multi-analytical (Py/GC-MS, FTIR, μ -Raman) approach, *Microchem. J.* 124 (2016) 929–939, <https://doi.org/10.1016/J.MICROC.2015.04.016>.
- [193] J. Zieba-Palus, R. Kowalski, The influence of the type of substrate on the possibility of spray paint identification for forensic purposes, *Vib. Spectrosc.* 95 (2018) 57–61, <https://doi.org/10.1016/J.VIBSPEC.2018.02.001>.
- [194] C. Muehlethaler, Y.P. Cheng, S.K. Islam, J.R. Lombardi, Contribution of Raman and surface enhanced Raman spectroscopy (SERS) to the analysis of vehicle headlights: Dye(s) characterization, *Forensic Sci. Int.* 287 (2018) 98–107, <https://doi.org/10.1016/J.FORSCIINT.2018.03.036>.
- [195] M.A. Malek, T. Nakazawa, H.W. Kang, K. Tsuji, C.U. Ro, Multi-Modal compositional analysis of layered paint chips of automobiles by the combined application of ATR-FTIR imaging, Raman microspectrometry, and SEM/EDX, *Molecules* 24 (2019), <https://doi.org/10.3390/MOLECULES24071381>.
- [196] S. Quevillon, N. Toupin, A. Tremblay, R. Fournier, C. Muehlethaler, Batch-to-batch variation in domestic paints: insights into the newly commercialized recycled paints, *Forensic Sci. Int.* 303 (2019), <https://doi.org/10.1016/J.FORSCIINT.2019.109946>.
- [197] M. Boscacci, S. Francone, K. Galli, S. Bruni, The brightest colors: a Fourier-transform Raman, surface-enhanced Raman, and thin-layer chromatography-surface-enhanced Raman spectroscopy study of fluorescent artists' paints, *J. Raman Spectrosc.* 51 (2020) 1108–1117, <https://doi.org/10.1002/JRS.5868>.
- [198] X. He, J. Wang, C. Gao, Y. Liu, Z. Li, N. Li, et al., Differentiation of White architectural paints by microscopic laser Raman spectroscopy and chemometrics, *Spectrochim. Acta A Mol. Biomol. Spectrosc.* 248 (2021) 119284, <https://doi.org/10.1016/J.SAA.2020.119284>.
- [199] M. Longoni, S. Bruni, Identification of synthetic organic pigments in contemporary Artists' paints by FT-IR and FT-Raman: an advanced analytical experiment, *J. Chem. Educ.* 98 (2021) 966–972, <https://doi.org/10.1021/ACS.JCHEMED.0C00875>.
- [200] R. Pause, I.D. van der Werf, K.J. van den Berg, Identification of pre-1950 synthetic organic pigments in artists' paints. A non-invasive approach using handheld Raman spectroscopy, *Heritage* 4 (2021) 1348–1365, <https://doi.org/10.3390/HERITAGE4030073>.
- [201] Affadu-Danful G.P.K. Application of Raman and infrared microscopy coupled with chemometrics for the forensic examination of automotive clear coats and paint smears 2022.
- [202] B. Borg, M. Dunn, A.S.M. Ang, C. Villis, Canvas painting analysis using spectroscopic analysis and microcharacterisation techniques, *Ital. Natl. Conf. Sens.* 22 (2022), <https://doi.org/10.3390/S22041442>.
- [203] V. Merk, W. Werncke, L. Pfeifer, Cross-section measurements of multilayer automotive paint samples using combined Raman spectroscopy and LIBS, *Analyst* 147 (2022) 5470–5476, <https://doi.org/10.1039/D2AN01474B>.
- [204] L. Chu, C. Guo, Q. Zhang, Q. Wang, Y. Ge, M. Hao, et al., Differentiation of multilayered automotive coatings with Fourier transform infrared spectroscopy, Raman spectroscopy and scanning electron microscope/energy dispersive x-ray spectrometer, *Pigment Resin Technol.* 53 (2022) 36–43, <https://doi.org/10.1108/PRT-04-2022-0047>.
- [205] V. Marazioti, A.M. Douvas, F. Katsaros, P. Koralli, C. Chochos, V.G. Gregoriou, et al., Chemical characterisation of artists' spray-paints: a diagnostic tool for urban art conservation, *Spectrochim. Acta A Mol. Biomol. Spectrosc.* 291 (2023) 122375, <https://doi.org/10.1016/J.SAA.2023.122375>.
- [206] A. Harshey, T. Das, V. Mishra, R.K. Shukla, A. Srivastava, Essentials of forensic physics, *Textb. Forensic Sci.* (2023) 377–423, https://doi.org/10.1007/978-981-99-1377-0_14.

- [207] L. Lei, G. Massonnet, Forensic analysis of White automotive paint of same manufacturer with Raman spectroscopy and chemometrics, *J. Raman Spectrosc.* 55 (2024) 148–160, <https://doi.org/10.1002/JRS.6626>.
- [208] R.A. Alblooshi, R.H. Alreimeithi, A.H. Aljannahi, A. Nahlé, Comparative forensic discrimination of pink lipsticks using Fourier transform infra-red and Raman spectroscopy, *Vib. Spectrosc.* 130 (2024), <https://doi.org/10.1016/J.VIBSPEC.2023.103640>.
- [209] M. Vieira, M.J. Melo, C. Conti, F. Pozzi, A combined approach to the vibrational characterization of medieval paints on parchment: handheld Raman spectroscopy and micro-SORS, *J. Raman Spectrosc.* 55 (2024) 116–124, <https://doi.org/10.1002/JRS.6632>.
- [210] R.W. Fitzpatrick, M.D. Raven, S.T. Forrester, A systematic approach to soil forensics: criminal case studies involving transference from crime scene to forensic evidence, *Crim. Environ. Soil Forensics* (2009) 105–127, https://doi.org/10.1007/978-1-4020-9204-6_8.
- [211] Gatta G.D., Mantovani L., Bromiley G.D. Raman Spectroscopy and Forensic Mineralogy 2023:141–169. https://doi.org/10.1007/978-3-031-08834-6_5.
- [212] Neha Yadav, Lav Kesharwani, Analytical instrument and its utilization in soil forensics: a review, *Indian J. Forensic Med. Toxicol.* 17 (2022) 48–52, <https://doi.org/10.37506/IJFMT.V17I1.18892>.
- [213] J. Jehlička, P. Vítek, H.G.M. Edwards, M. Hargreaves, T. Čapoun, Rapid outdoor non-destructive detection of organic minerals using a portable Raman spectrometer, *J. Raman Spectrosc.* 40 (2009) 1645–1651, <https://doi.org/10.1002/JRS.2313>.
- [214] J. Jehlička, H.G.M. Edwards, A. Culka, Using portable Raman spectrometers for the identification of organic compounds at low temperatures and high altitudes: exobiological applications, *Philos. Trans. R. Soc. A Math. Phys. Eng. Sci.* 368 (2010) 3109–3125, <https://doi.org/10.1098/STA.2010.0075>.
- [215] Alajtal A.I. Raman spectroscopic application for the analysis of organic compounds and minerals of astrobiological significance. The detection and discrimination of organic compounds and mineral analogues in pure and mixed samples of astrobiological significance using raman spectroscopy, XRD and scanning electron microscopy 2010.
- [216] H. Edwards, T. Munshi, I. Scowen, A. Surtees, G.T. Swindles, Development of oxidative sample preparation for the analysis of forensic soil samples with near-IR Raman spectroscopy, *J. Raman Spectrosc.* 43 (2012) 323–325, <https://doi.org/10.1002/JRS.3031>.
- [217] A. Surtees, Development of geochemical identification and discrimination by Raman spectroscopy, *Dev. Raman Spectrosc. Methods Appl. whole Soil Anal. Sep. Volcan. ashes tephrochronology* (2015).
- [218] E.M.J. Schotsmans, A.S. Wilson, R. Brettell, T. Munshi, H.G.M. Edwards, Raman spectroscopy as a non-destructive screening technique for studying White substances from archaeological and forensic burial contexts, *J. Raman Spectrosc.* 45 (2014) 1301–1308, <https://doi.org/10.1002/JRS.4526>.
- [219] B.W. Kammrath, A. Koutrakos, J. Castillo, C. Langley, D. Huck-Jones, Morphologically-directed Raman spectroscopy for forensic soil analysis, *Forensic Sci. Int.* 285 (2018) e25–e33, <https://doi.org/10.1016/J.FORSCIINT.2017.12.034>.
- [220] P.M. Nkebiwe, K. Sowoidnich, M. Maiwald, B. Sumpf, T.E. Hartmann, D. Wanke, et al., Detection of calcium phosphate species in soil by confocal μ -Raman spectroscopy, *J. Plant Nutr. Soil Sci.* 185 (2022) 221–231, <https://doi.org/10.1002/JPLN.202100233>.
- [221] Z. Xing, C. Du, Y. Shen, F. Ma, J. Zhou, A method combining FTIR-ATR and Raman spectroscopy to determine soil organic matter: improvement of prediction accuracy using competitive adaptive reweighted sampling (CARS), *Comput. Electron Agric.* 191 (2021), <https://doi.org/10.1016/J.COMPAE.2021.106549>.
- [222] Y. Bai, W. Yang, Z. Wang, Y. Cao, M. Li, Improving the estimation accuracy of soil organic matter based on the fusion of near-infrared and Raman spectroscopy using the outer-product analysis, *Comput. Electron Agric.* 219 (2024) 108760, <https://doi.org/10.1016/J.COMPAE.2024.108760>.
- [223] Pitts K.M. The Use of Raman Spectroscopy and X-ray Diffraction for the Examination of Associated Clay and Silt Coatings of Forensically Important Sandy Soils from the Swan Coastal Plain, Western Australia 2022.
- [224] T.G. Newland, K. Pitts, S.W. Lewis, Negative result: application of Raman spectroscopy to the forensic analysis of an arid, sandy, soil, *Forensic Science International Reports* 8 (2023) 100343, <https://doi.org/10.1016/J.FSIR.2023.100343>.
- [225] E.A. Stefaniak, A. Alsecc, I.E. Sajó, A. Worobiec, Z. Máthé, S. Török, et al., Recognition of uranium oxides in soil particulate matter by means of μ -Raman spectrometry, *J. Nucl. Mater.* 381 (2008) 278–283, <https://doi.org/10.1016/J.JNUCMAT.2008.08.036>.
- [226] G. Corrado, S. Sanchez-Cortes, O. Francioso, J.V. Garcia-Ramos, Surface-enhanced Raman and fluorescence joint analysis of soil humic acids, *Anal. Chim. Acta* 616 (2008) 69–77, <https://doi.org/10.1016/J.ACA.2008.04.019>.
- [227] L. Bellot-Gurlet, D. Neff, S. Réguer, J. Monnier, M. Saheb, P. Dillmann, Raman studies of corrosion layers formed on archaeological irons in various media, *J. Nano Res.* 8 (2009) 147–156, <https://doi.org/10.4028/WWW.SCIENTIFIC.NET/JNANOR.8.147>.
- [228] Z.C. Ling, A. Wang, B.L. Jolliffe, Mineralogy and geochemistry of four lunar soils by laser-Raman study, *Icarus* 211 (2011) 101–113, <https://doi.org/10.1016/J.ICARUS.2010.08.020>.
- [229] M.L. Roldán, G. Corrado, O. Francioso, S. Sanchez-Cortes, Interaction of soil humic acids with herbicide paraquat analyzed by surface-enhanced Raman scattering and fluorescence spectroscopy on silver plasmonic nanoparticles, *Anal. Chim. Acta* 699 (2011) 87–95, <https://doi.org/10.1016/J.ACA.2011.05.001>.
- [230] R.L. Frost, S.J. Palmer, Raman spectroscopic study of the minerals diadochite and destinezite $\text{Fe}_3\text{+2}(\text{PO}_4,\text{SO}_4)_2(\text{OH}) \cdot 6\text{H}_2\text{O}$: implications for soil science, *J. Raman Spectrosc.* 42 (2011) 1589–1595, <https://doi.org/10.1002/JRS.2883>.
- [231] F. Kizewski, Y.-T. Liu, A. Morris, D. Hesterberg, Spectroscopic approaches for phosphorus speciation in soils and other environmental systems, *J. Environ. Qual.* 40 (2011) 751–766, <https://doi.org/10.2134/JEQ2010.0169>.
- [232] L. Zheng, W.S. Lee, M. Li, A. Katti, C. Yang, H. Li, et al., Analysis of soil phosphorus concentration based on Raman spectroscopy. Multispectral, hyperspectral, and ultraspectral remote sensing technology, *Tech. Appl. IV* 8527 (2012) 852718, <https://doi.org/10.1117/12.977436>.
- [233] J.A. Carrero, G. Arana, J. Madariaga, Important Use Raman Spectrosc. Help Under Impact Traffic Roadside Soils Plants (2012).
- [234] J. Ribeiro-Soares, L.G. Cançado, N.P.S. Falcão, E.H. Martins Ferreira, C.A. Achete, A. Jorio, The use of Raman spectroscopy to characterize the carbon materials found in amazonian anthrosols, *J. Raman Spectrosc.* 44 (2013) 283–289, <https://doi.org/10.1002/JRS.4191>.
- [235] J.A. Carrero, N. Goienaga, M. Olivares, I. Martinez-Arkarazo, G. Arana, J. M. Madariaga, Raman spectroscopy assisted with XRF and chemical simulation to assess the synergic impacts of guardrails and traffic pollutants on urban soils, *J. Raman Spectrosc.* 43 (2012) 1498–1503, <https://doi.org/10.1002/JRS.4089>.
- [236] N. Taquet, J. Pironon, P. De Donato, H. Lucas, O. Barres, Efficiency of combined FTIR and Raman spectrometry for online quantification of soil gases: application to the monitoring of carbon dioxide storage sites, *Int. J. Greenh. Gas. Control* 12 (2013) 359–371, <https://doi.org/10.1016/J.IJGGC.2012.10.003>.
- [237] S.J. Parikh, K.W. Goyne, A.J. Margenot, F.N.D. Mukome, F.J. Calderón, Soil chemical insights provided through vibrational spectroscopy, *Adv. Agron.* 126 (2014) 1–148, <https://doi.org/10.1016/B978-0-12-800132-5.00001-8>.
- [238] A.S. Luna, I.C.A. Lima, W.F.C. Rocha, J.R. Araújo, A. Kuznetsov, E.H.M. Ferreira, et al., Classification of soil samples based on Raman spectroscopy and X-ray fluorescence spectrometry combined with chemometric methods and variable selection, *Anal. Methods* 6 (2014) 8930–8939, <https://doi.org/10.1039/C4AY01967A>.
- [239] J. Wei, A. Wang, J.L. Lambert, D. Wettergreen, N. Cabrol, K. Warren-Rhodes, et al., Autonomous soil analysis by the Mars Micro-beam Raman spectrometer (MMRS) on-board a rover in the atacama desert: a terrestrial test for planetary exploration, *J. Raman Spectrosc.* 46 (2015) 810–821, <https://doi.org/10.1002/JRS.4656>.
- [240] Z. Xing, C. Du, Y. Zeng, F. Ma, J. Zhou, Characterizing typical farmland soils in China using Raman spectroscopy, *Geoderma* 268 (2016) 147–155, <https://doi.org/10.1016/J.GEODERMA.2016.01.029>.
- [241] J. Inoue, A. Yoshie, T. Tanaka, T. Onji, Y. Inoue, Disappearance and alteration process of charcoal fragments in cumulative soils studied using Raman spectroscopy, *Geoderma* 285 (2017) 164–172, <https://doi.org/10.1016/J.GEODERMA.2016.09.032>.
- [242] P. Nie, T. Dong, S. Xiao, L. Lin, Y. He, F. Qu, Quantitative determination of thiabendazole in soil extracts by Surface-Enhanced Raman spectroscopy, *Mol.* 2018 Vol. 23 Page 1949 23 (2018) 1949, <https://doi.org/10.3390/MOLECULES23081949>.
- [243] Y. He, S. Xiao, T. Dong, P. Nie, Gold nanoparticles for qualitative detection of deltamethrin and carbofuran residues in soil by surface enhanced Raman scattering (SERS), *Int. J. Mol. Sci.* 20 (2019) 1731, <https://doi.org/10.3390/IJMS20071731>.
- [244] L.S. Theurer, M. Maiwald, B. Sumpf, Shifted excitation Raman difference spectroscopy: a promising tool for the investigation of soil, *Eur. J. Soil Sci.* 72 (2021) 120–124, <https://doi.org/10.1111/EJSS.12928>.
- [245] W. Cao, Y. Luo, J. Li, A. Qian, Q. Wang, X. Wang, et al., Detection of benzo[a]pyrene with silver nanorod substrate in river water and soil based on surface-enhanced Raman scattering, *Results Chem.* 3 (2021) 100126, <https://doi.org/10.1016/J.RECHEM.2021.100126>.
- [246] V. Sharma, R. Chauhan, R. Kumar, Spectral characteristics of organic soil matter: a comprehensive review, *Microchem. J.* 171 (2021) 106836, <https://doi.org/10.1016/J.MICROC.2021.106836>.
- [247] M. Maiwald, K. Sowoidnich, B. Sumpf, Portable shifted excitation Raman difference spectroscopy for on-site soil analysis, *J. Raman Spectrosc.* 53 (2022) 1560–1570, <https://doi.org/10.1002/JRS.6400>.
- [248] K. Sowoidnich, S. Vogel, M. Maiwald, B. Sumpf, Determination of soil constituents using shifted excitation Raman difference spectroscopy, *Appl. Spectrosc.* 76 (2022) 712–722, <https://doi.org/10.1177/00037028211064907>.
- [249] L. Pan, P. Zhang, C. Daengngam, S. Peng, M. Chongcheawchamnan, A review of artificial intelligence methods combined with Raman spectroscopy to identify the composition of substances, *J. Raman Spectrosc.* 53 (2022) 6–19, <https://doi.org/10.1002/JRS.6225>.
- [250] J. Chen, M. Liu, H. Yuan, S. Huang, J. Zhao, J. Tao, et al., Surface-enhanced Raman spectroscopy for classification of testosterone propionate and nandrolone residues in chicken, *Vib. Spectrosc.* 99 (2018) 7–12, <https://doi.org/10.1016/J.VIBSPEC.2018.08.020>.
- [251] Y. Dai, Z. Liu, S. Zhao, Fusion of Laser-Induced breakdown spectroscopy and Raman spectroscopy for mineral identification based on machine learning, *Molecules* 29 (2024), <https://doi.org/10.3390/MOLECULES29143317/S1>.
- [252] J. Xue, H. Yue, W. Lu, Y. Li, G. Huang, Y.V. Fu, Application of Raman spectroscopy and machine learning for candida auris identification and characterization, *Appl. Environ. Microbiol.* 90 (2024) e0102524, <https://doi.org/10.1128/AEM.01025-24>.
- [253] Y.V. Kistenev, A.V. Borisov, A.A. Samarinova, S. Colón-Rodríguez, I.K. Lednev, A novel Raman spectroscopic method for detecting traces of blood on an

- interfering substrate, *Sci. Rep.* 2023 131 13 (2023) 1–15, <https://doi.org/10.1038/s41598-023-31918-9>.
- [254] A.V. Borisov, M.S. Snegerev, S. Colón-Rodríguez, M.A. Fikiel, I.K. Lednev, Y. V. Kistenev, Identification of semen traces at a crime scene through Raman spectroscopy and machine learning, *Sci. Rep.* 14 (2024) 23070, <https://doi.org/10.1038/S41598-024-73563-W>.
- [255] T.Y. Huang, J. Wang, Q. Liu, J. Yu, The application of wavelet transform of Raman spectra to facilitate transfer learning for gasoline detection and classification, *Talanta Open* 5 (2022) 100106, <https://doi.org/10.1016/J.TALO.2022.100106>.
- [256] J. Grijalva, T.Y. Huang, J. Yu, P. Buzzini, D. Williams, J.T. Davidson, et al., Analysis of major cannabinoids using Raman microscopy, density functional theory, chemometrics and a novel artificial intelligence approach, *Talanta Open* 10 (2024) 100337, <https://doi.org/10.1016/J.TALO.2024.100337>.